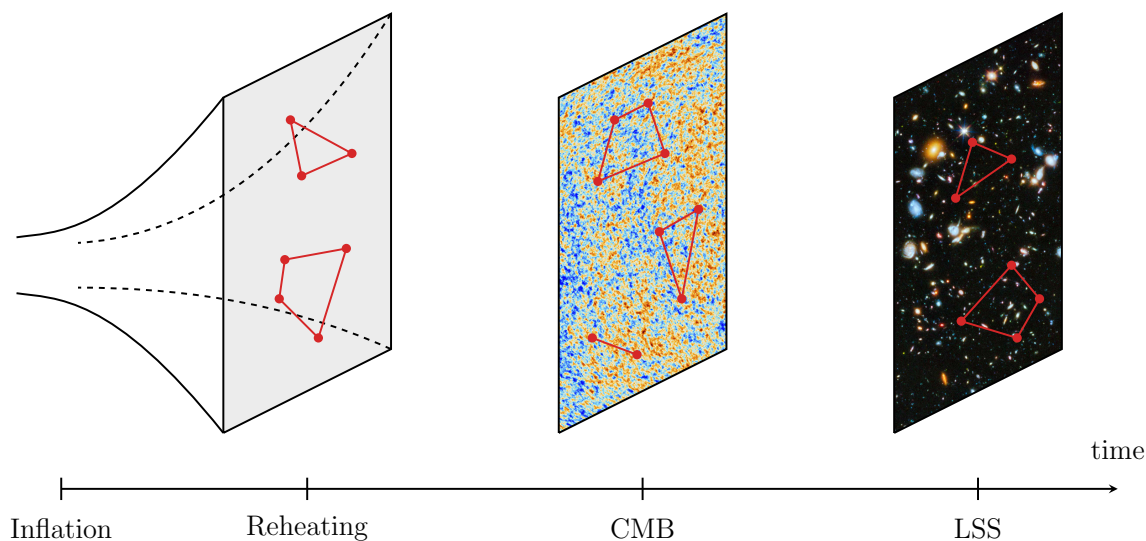


Lectures on Cosmological Correlations

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OUTLINE

- I. Motivation
 - II. Primordial Statistics
 - III. Inflationary Fluctuations
 - VI. In-In Formalism
 - V. Wavefunction Approach
 - VI. Cosmological Bootstrap
 - VII. Outlook
- Cosmological
Collider Physics

Lecture notes and lecture scripts can be found at:

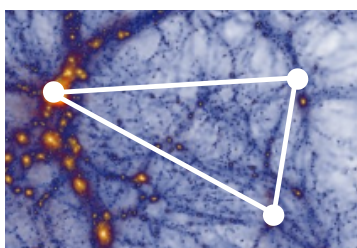
<https://github.com/ddbaumann/cosmo-correlators/Cargese-Lectures.pdf>

I. MOTIVATION

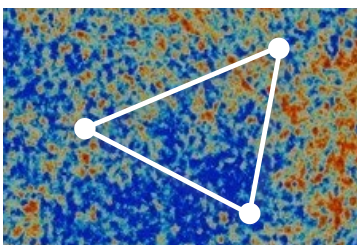
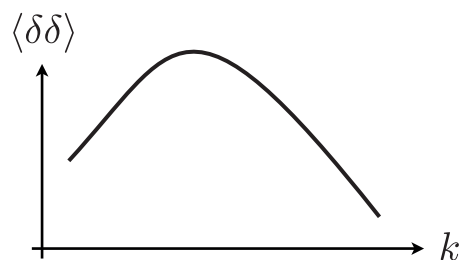
One of the biggest questions in all of science is the origin of structure in the Universe. What created everything that we see around us? An important clue lies in the fact that the structures in the Universe aren't distributed randomly, but displays large-scale correlations. These correlations are a fossil record of the early universe, and by measuring them we hope to uncover how the cosmological perturbations formed and evolved. In this course, we will give an introduction to these "cosmological correlations".

1.1 Practical Motivation

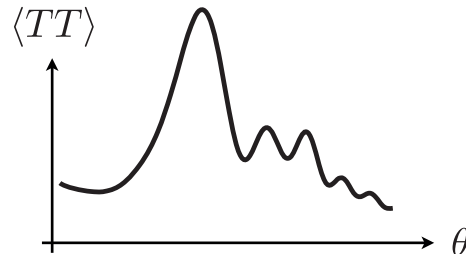
In cosmology, we measure **spatial correlations**:



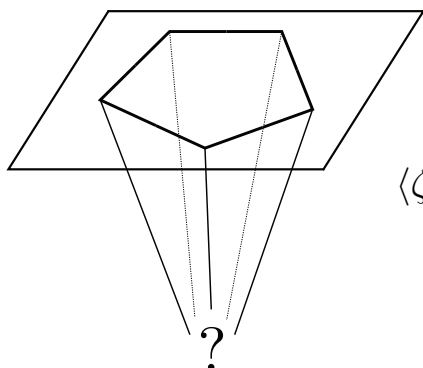
$$\langle \delta\rho(\mathbf{x}_1) \cdots \delta\rho(\mathbf{x}_N) \rangle$$



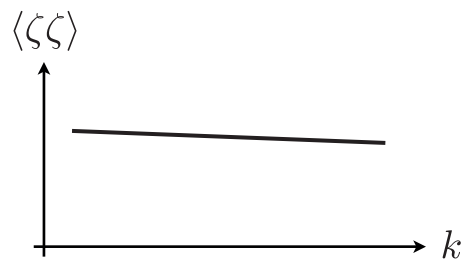
$$\langle \delta T(\theta_1) \cdots \delta T(\theta_N) \rangle$$



These correlations can be traced back to the origin of the hot Big Bang:



$$\langle \zeta(\mathbf{x}_1) \cdots \zeta(\mathbf{x}_N) \rangle$$



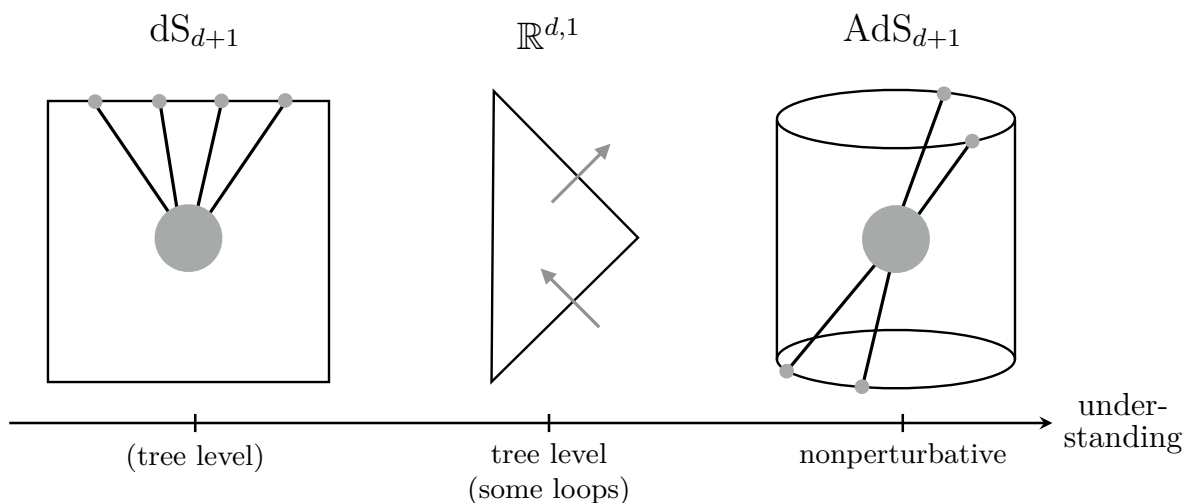
Where did the primordial correlations come from?

- Clue 1: The correlations span superhorizon scales.
- Clue 2: They are scale-invariant.

This suggests that the fluctuations were created **before the hot Big Bang**, during a phase of approximate time-translation invariance (= **inflation**).

1.2 Conceptual Motivation

The study of cosmological correlators is also of conceptual interest:



Our understanding of quantum field theory in de Sitter space (cosmology) is still rather underdeveloped $\Rightarrow$ Opportunity for you to make progress!

II. PRIMORDIAL STATISTICS

In cosmology, we only predict the statistical properties of the Universe. This means that we can't predict whether a galaxy will be found at a specific position $\mathbf{x}$. Instead, our theories can only tell us about the probability $P(\mathbf{x})$ of finding a galaxy at $\mathbf{x}$, or the conditional probability $P(\mathbf{x}_2, \mathbf{x}_1)$ that a second galaxy is at $\mathbf{x}_2$ given a galaxy at $\mathbf{x}_1$. Using these probabilities, we can derive spatial correlation functions, which are the main observables in cosmology.

2.1 Random Fields

Cosmological fluctuations $(\delta\rho, \delta g_{\mu\nu}, \dots)$ are “random fields” $\equiv \Phi(t, \mathbf{x})$.

- *Correlation functions:*

$$\langle \Phi_1 \Phi_2 \dots \Phi_N \rangle \equiv \int \Phi_1 \Phi_2 \dots \Phi_N P(\Phi_1, \dots, \Phi_N) \mathcal{D}\Phi_1 \dots \mathcal{D}\Phi_N,$$

where $\Phi_i \equiv \Phi(t, \mathbf{x}_i)$.

- *Ergodic hypothesis:*

Average over statistical ensemble = spatial average over one realization.

$$\langle \Phi(t, \mathbf{x}) \rangle = \int \Phi P(\Phi) d\Phi = \underset{\substack{\uparrow \\ \text{ergodicity}}}{\frac{1}{V} \int d^3x \Phi(t, \mathbf{x})} \equiv \bar{\Phi}(t).$$

- *Symmetries:*

The correlations are statistically homogeneous and isotropic:

$$\langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \rangle = \underset{\substack{\uparrow \\ \text{homogeneity}}}{\xi_\Phi(\mathbf{x}_1 - \mathbf{x}_2)} = \underset{\substack{\uparrow \\ \text{isotropy}}}{\xi_\Phi(|\mathbf{x}_1 - \mathbf{x}_2|)} \equiv \xi_\Phi(x_{12}),$$

- *Fourier space:*

Reasons cosmologists like Fourier space:

- 1) Fourier modes “decouple” (in linear theory)
- 2) Correlators “diagonalize”

Fourier expansion of the field is

$$\Phi(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{x}} \Phi_{\mathbf{k}}(t)$$

$\uparrow$
 complex : $\Phi_{\mathbf{k}}^* = \Phi_{-\mathbf{k}}$

$\Rightarrow$ Equal-time correlators: $\langle \Phi_{\mathbf{k}_1} \Phi_{\mathbf{k}_2} \cdots \Phi_{\mathbf{k}_N} \rangle$.

$\Rightarrow$ Two-point function:

$$\begin{aligned} \langle \Phi_{\mathbf{k}_1} \Phi_{\mathbf{k}_2} \rangle &= \int d^3x_1 d^3x_2 e^{-i\mathbf{k}_1\cdot\mathbf{x}_1} e^{-i\mathbf{k}_2\cdot\mathbf{x}_2} \langle \Phi(\mathbf{x}_1) \Phi(\mathbf{x}_2) \rangle \\ &= \underbrace{\int d^3x_2 e^{-i(\mathbf{k}_1+\mathbf{k}_2)\cdot\mathbf{x}_2}}_{(2\pi)^3\delta(\mathbf{k}_1+\mathbf{k}_2)} \underbrace{\int d^3x_{12} e^{-i\mathbf{k}_1\cdot\mathbf{x}_{12}} \xi_{\Phi}(x_{12})}_{P_{\Phi}(k_1)} \end{aligned}$$

$\uparrow$ $\uparrow$
 homogeneity power spectrum

- *Power spectrum:*

$$\langle \Phi_{\mathbf{x}_1} \Phi_{\mathbf{x}_2} \rangle \equiv \xi_{\Phi}(|\mathbf{x}_1 - \mathbf{x}_2|) \quad \Rightarrow \quad \langle \Phi_{\mathbf{k}_1} \Phi_{\mathbf{k}_2} \rangle = (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2) P_{\Phi}(k_1),$$

where

$$\begin{aligned} P_{\Phi}(k) &= \int d^3x e^{-i\mathbf{k}\cdot\mathbf{x}} \xi_{\Phi}(x), \\ \xi_{\Phi}(0) &= \int \frac{d^3k}{(2\pi)^3} P_{\Phi}(k) = \int d \ln k \frac{k^3}{2\pi^2} P_{\Phi}(k) \equiv \int d \ln k \Delta_{\Phi}^2(k) \end{aligned}$$

$\uparrow$ $\uparrow$
 variance dimensionless
 power spectrum

The spectrum is *scale-invariant* if $\Delta_{\Phi}^2(k)$ is a constant.

- For *Gaussian random fields*, all correlators are functions of $P_{\Phi}(k)$.
 $\Rightarrow$ “connected correlators” vanish for $N > 2$.

2.2 Cosmological Perturbations

Write the metric and stress-energy tensor as

$$\begin{aligned} g_{\mu\nu}(t, \mathbf{x}) &= \bar{g}_{\mu\nu}(t) + \delta g_{\mu\nu}(t, \mathbf{x}) , \\ T_{\mu\nu}(t, \mathbf{x}) &= \bar{T}_{\mu\nu}(t) + \delta T_{\mu\nu}(t, \mathbf{x}) , \end{aligned}$$

and then expand $\nabla^\mu T_{\mu\nu} = 0$ and $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ to linear order.

- The form of the perturbations depends on the choice of coordinates.
- In the *comoving gauge* ($T_{0i} \equiv 0$), we can write the spatial metric as

$$g_{ij} = a^2 e^{2\zeta(t, \mathbf{x})} \delta_{ij} ,$$

where ζ is the *comoving curvature perturbation*: $a^2 R_{(3)} = 4\nabla^2 \zeta$.

- During inflation:

$$\zeta = \left(-\frac{H}{\dot{\phi}} \right) \delta\phi ,$$

where $\delta\phi$ is the inflaton perturbation in spatially flat gauge.

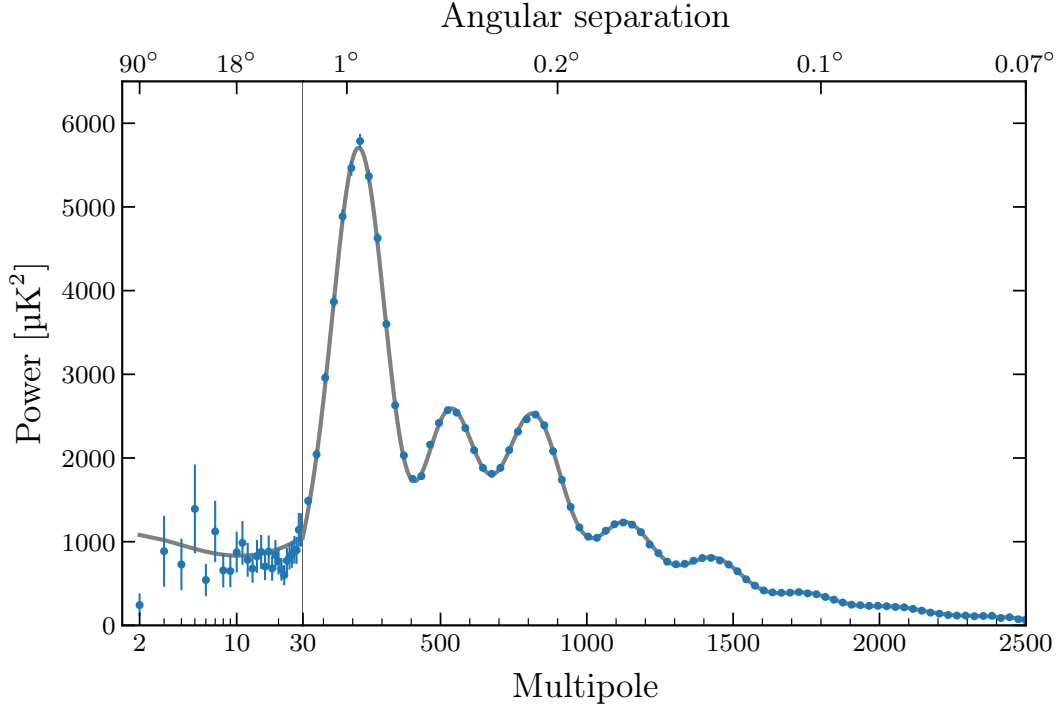
- After inflation: $\zeta = \text{const}$ for $k < (aH)$ (superhorizon scales).
- Each observable $\mathcal{O}$ is related to $\zeta_{\mathbf{k}}(t_i)$ by a *transfer function*:

$$\mathcal{O}(\mathbf{x}, t) = \int \frac{d^3k}{(2\pi)^3} T_{\mathcal{O}}(k; t, t_i) \zeta_{\mathbf{k}}(t_i) e^{i\mathbf{k}\cdot\mathbf{x}} .$$

$\Rightarrow$ Fluctuations can be traced back to $\zeta_{\mathbf{k}}(t_i)$.

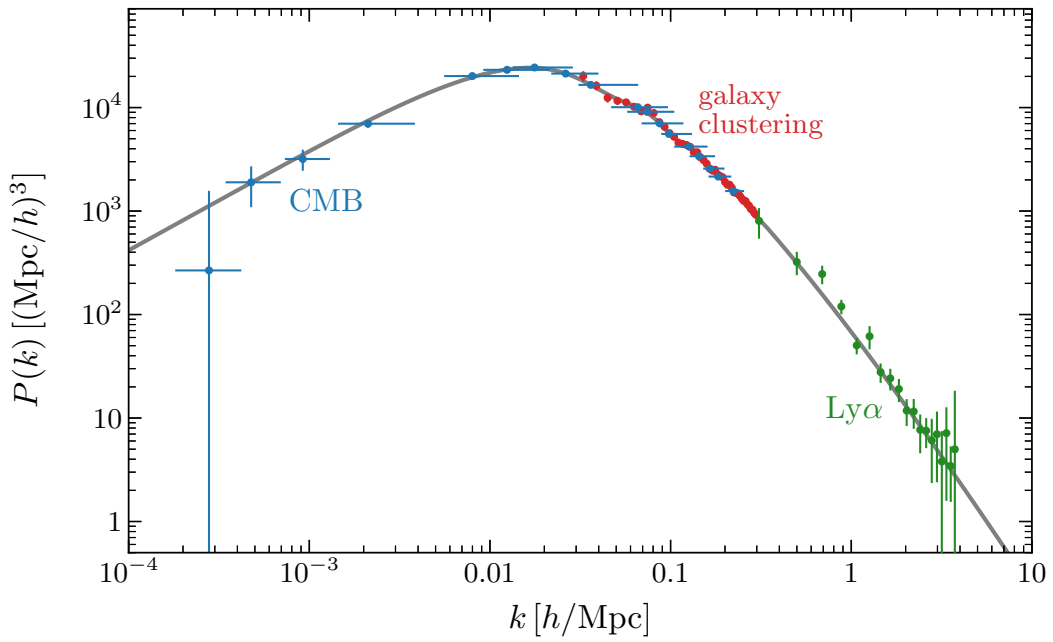
$\Rightarrow$ Physics before the hot Big Bang ($t < t_i$) is encoded in $\langle \zeta_{\mathbf{k}_1} \cdots \zeta_{\mathbf{k}_N} \rangle$.

2.3 Observed Correlations



The shape of the CMB power spectrum depends on

$$\{ H_0, \Omega_K, \Omega_\Lambda, \Omega_m, \Omega_b, \Omega_\nu, A_s e^{-2\tau}, n_s \}.$$



The shape of the matter power spectrum depends on $\{ \Omega_m, \sigma_8 \}$.

2.4 Initial Conditions

We have learned **four important facts** about the initial conditions:

1. *Superhorizon*

The fluctuations spanned superhorizon scales at photon decoupling.
⇒ They must have been created before the hot Big Bang.

2. *Scale-invariant*

The primordial fluctuations were scale invariant.
⇒ This is a natural prediction of inflation.

3. *Adiabatic*

The fluctuations were adiabatic (i.e. no fluctuations in composition).
⇒ All fluctuations are sourced by a single scalar mode:

$$\zeta(\mathbf{x}) = \frac{1}{3} \left(\frac{\delta\rho}{\bar{\rho} + \bar{P}} \right)_i, \quad i = \text{dark matter, baryons, photons}$$

4. *Gaussian*

The primordial fluctuations were highly Gaussian:

$$\frac{\langle \zeta \zeta \zeta \rangle}{\langle \zeta \zeta \rangle^{3/2}} < 0.1\%$$

⇒ This a natural prediction for quantum fluctuations of free fields.
⇒ A lot of the physics of inflation is encoded in non-Gaussian correlations.

III. INFLATIONARY FLUCTUATIONS

One of the most remarkable features of inflation is that it provides a natural mechanism for creating the primordial density fluctuations that seeded the structure in the Universe. In this chapter, we will derive the spectrum of quantum fluctuations produced during inflation.

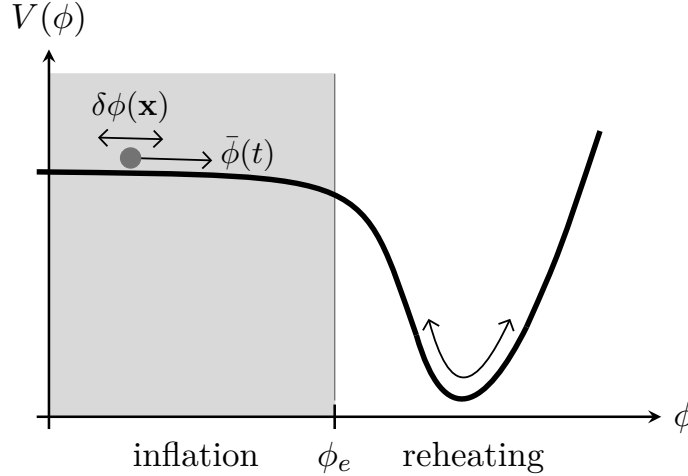
3.1 Basics of Inflation

Consider the action

$$S = \int d^4x \sqrt{-g} \left(\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right),$$

where $M_{\text{Pl}} \equiv (8\pi G)^{-1/2}$.

- Leads to accelerated expansion ($\ddot{a} > 0$) when the potential $V(\phi)$ is flat.
- Inflation ends when the potential steepens.
- Quantum fluctuations in the field create density perturbations.



Varying the inflaton action with respect to the metric gives

$$T_{\mu\nu} \equiv -\frac{2}{\sqrt{-g}} \frac{\delta S_\phi}{\delta g^{\mu\nu}} = \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - V(\phi) \right).$$

- Assuming $\phi = \phi(t)$ for the homogeneous background, we get

$$\rho_\phi = -T^0_0 = \frac{1}{2}\dot{\phi}^2 + V(\phi).$$

- The Friedmann equation becomes

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{\rho_\phi}{3M_{\text{Pl}}^2} = \frac{1}{3M_{\text{Pl}}^2} \left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right).$$

- Slow-roll inflation occurs when $\frac{1}{2}\dot{\phi}^2 \ll V$.
 - During inflation, $H \approx \text{const} \Rightarrow a(t) \approx e^{Ht}$.
-

Varying the inflaton action with respect to the field gives

$$\ddot{\phi} + 3H\dot{\phi} = -\frac{dV}{d\phi}.$$

- Sustained slow-roll inflation occurs when $\ddot{\phi} \ll 3H\dot{\phi}$.
-

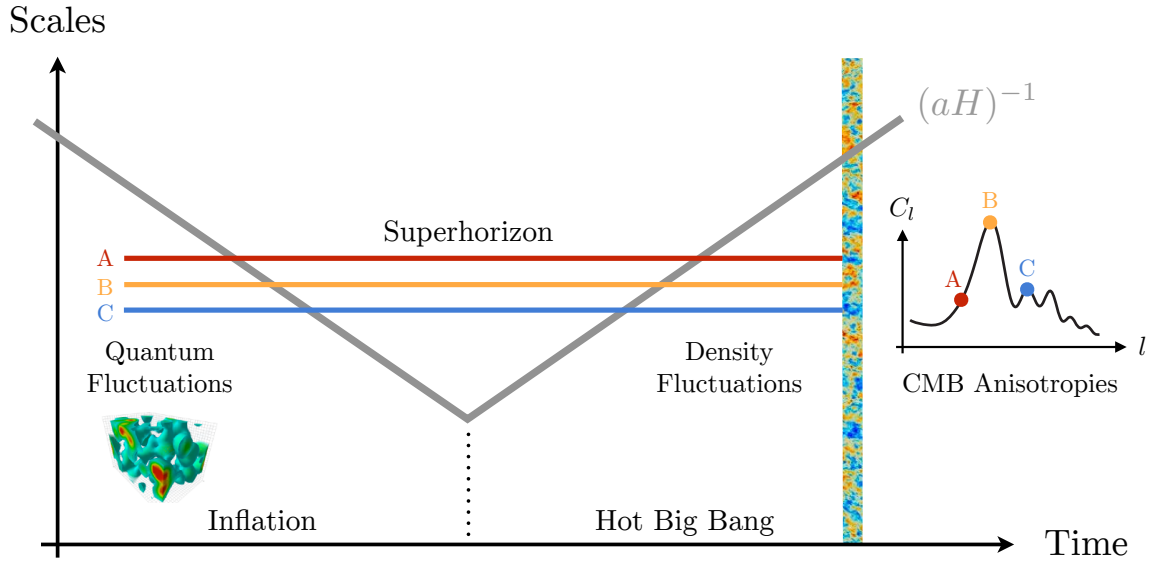
Accelerated expansion implies a shrinking comoving Hubble radius:

$$\frac{d}{dt}(aH)^{-1} = \frac{d}{dt}(\dot{a})^{-1} = -\frac{\ddot{a}}{(\dot{a})^2} < 0.$$

This has important consequences for the evolution of perturbations:

- Small-scale perturbations are stretched to large scales.
- Quantum fluctuations become classical perturbations.
- Perturbations are frozen on super-Hubble scales.

See diagram on the next page.



The shrinking Hubble radius can also be written as

$$\frac{d}{dt}(aH)^{-1} = -\frac{\dot{a}H + a\dot{H}}{(aH)^2} = -\frac{1}{a}(1 - \varepsilon),$$

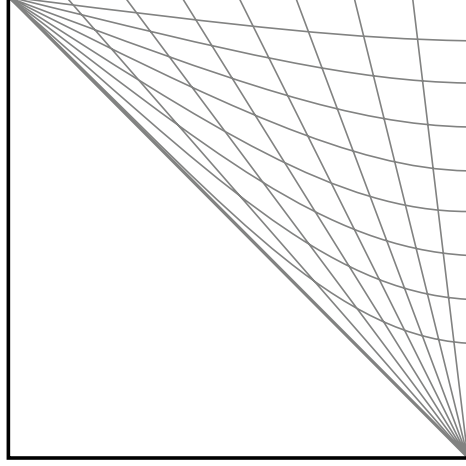
where the “slow-roll parameter” is

$$\varepsilon \equiv -\frac{\dot{H}}{H^2}.$$

- Inflation requires $\varepsilon < 1$.
- Scale-invariant perturbations require $\varepsilon \ll 1$.

3.2 Free Scalars in De Sitter

The spacetime during inflation is approximately **de Sitter space**.



In conformal time, $d\eta = dt/a(t)$, the de Sitter metric is

$$ds^2 = \frac{-d\eta^2 + d\mathbf{x}^2}{(H\eta)^2},$$

where $-\infty < \eta < 0$. The primordial correlations live on the **future boundary** at $\eta_* \approx 0$. We consider a free, massless scalar in this background.

Classical dynamics

The action is

$$\begin{aligned} S &= \frac{1}{2} \int d\eta d^3x a^2 \left[(\phi')^2 - (\nabla\phi)^2 \right] \\ &= \frac{1}{2} \int d\eta d^3x \left[(u')^2 - (\nabla u)^2 + \frac{a''}{a} u^2 \right], \end{aligned}$$

where $u \equiv a(\eta)\phi$.

- The classical equation of motion then is

$$u'' - \nabla^2 u - \frac{a''}{a} u = 0 \quad \Longrightarrow \quad \boxed{u''_{\mathbf{k}} + \left(k^2 - \frac{2}{\eta^2} \right) u_{\mathbf{k}} = 0}.$$

- At early times ($|k\eta| \gg 1$):

$$u''_{\mathbf{k}} + k^2 u_{\mathbf{k}} = 0 \quad \Longrightarrow \quad u_{\mathbf{k}} = \frac{1}{\sqrt{2k}} e^{\pm i k \eta}.$$

- At late times ($|k\eta| \ll 1$):

$$u_{\mathbf{k}}'' - \frac{2}{\eta^2} u_{\mathbf{k}} = 0 \quad \Longrightarrow \quad u_{\mathbf{k}} = A_{\mathbf{k}} \eta^{-1} + B_{\mathbf{k}} \eta^2.$$

- Most general solution:

$$u_{\mathbf{k}}(\eta) = C_{\mathbf{k}} \left(1 - \frac{i}{k\eta}\right) \frac{e^{-ik\eta}}{\sqrt{2k}} + D_{\mathbf{k}} \left(1 + \frac{i}{k\eta}\right) \frac{e^{ik\eta}}{\sqrt{2k}}.$$

Canonical quantization

- Introduce **operators** $\hat{u}, \hat{\pi} = \partial_\eta \hat{u}$.
- Impose **commutation relations**:

$$[\hat{u}(\eta, \mathbf{x}), \hat{\pi}(\eta, \mathbf{x}')] = i\delta(\mathbf{x} - \mathbf{x}') \quad \Longrightarrow \quad [\hat{u}_{\mathbf{k}}(\eta), \hat{\pi}_{\mathbf{k}'}(\eta)] = i(2\pi)^3 \delta(\mathbf{k} + \mathbf{k}').$$

- Define **mode expansion**:

$$\hat{u}(\eta, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \left(u_k(\eta) \hat{a}_{\mathbf{k}} + u_k^*(\eta) \hat{a}_{-\mathbf{k}}^\dagger \right) e^{i\mathbf{k} \cdot \mathbf{x}},$$

where

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta(\mathbf{k} - \mathbf{k}') \quad \Longrightarrow \quad u_k(\eta) \partial_\eta u_k^*(\eta) - u_k^*(\eta) \partial_\eta u_k(\eta) = i.$$

- Define **vacuum**:

$$\hat{a}_{\mathbf{k}}|0\rangle = 0.$$

Bunch–Davies vacuum

Choose the minimum energy state at early times:

$$\lim_{k\eta \rightarrow -\infty} u_k(\eta) = \frac{1}{\sqrt{2k}} e^{-ik\eta} \quad \Longrightarrow \quad u_k(\eta) = \left(1 - \frac{i}{k\eta}\right) \frac{e^{-ik\eta}}{\sqrt{2k}}.$$

Zero-point fluctuations

The variance of the field operator is

$$\begin{aligned}
\langle |\hat{u}|^2 \rangle &\equiv \langle 0 | \hat{u}(\eta, \mathbf{0}) \hat{u}(\eta, \mathbf{0}) | 0 \rangle \\
&= \int \frac{d^3 k}{(2\pi)^3} \int \frac{d^3 k'}{(2\pi)^3} \langle 0 | \overline{(u_k^*(\eta) \hat{a}_{-\mathbf{k}}^\dagger + u_k(\eta) \hat{a}_{\mathbf{k}})} (u_{k'}(\eta) \hat{a}_{\mathbf{k}'} + \overline{u_{k'}^*(\eta) \hat{a}_{-\mathbf{k}'}^\dagger}) | 0 \rangle \\
&= \int \frac{d^3 k}{(2\pi)^3} \int \frac{d^3 k'}{(2\pi)^3} u_k(\eta) u_{k'}^*(\eta) \langle 0 | [\hat{a}_{\mathbf{k}}, \hat{a}_{-\mathbf{k}'}^\dagger] | 0 \rangle \\
&= \int \frac{d^3 k}{(2\pi)^3} |u_k(\eta)|^2 \\
&= \int d \log k \frac{k^3}{2\pi^2} |u_k(\eta)|^2 .
\end{aligned}$$

We define the (dimensionless) **power spectrum** as

$$\Delta_u^2(k, \eta) \equiv \frac{k^3}{2\pi^2} |u_k(\eta)|^2 .$$

Substituting the Bunch–Davies mode function, we find

$$\Delta_\phi^2(k, \eta) = \frac{\Delta_u^2(k, \eta)}{a^2(\eta)} = \left(\frac{H}{2\pi} \right)^2 \left[1 + (k\eta)^2 \right] \xrightarrow{k\eta \rightarrow 0} \boxed{\left(\frac{H}{2\pi} \right)^2} .$$

Since $H \approx \text{const}$, the spectrum is **scale-invariant**.

3.3 Curvature Perturbations

Now consider the coupled inflaton-metric fluctuations.

- ADM metric:

$$ds^2 = -N^2 dt^2 + h_{ij}(N^i dt + dx^i)(N^j dt + dx^j),$$

so that

$$g_{\mu\nu} = \left(\frac{-N^2 + N_i N^i}{N_i} \middle| \frac{N_j}{h_{ij}} \right), \quad g^{\mu\nu} = \frac{1}{N^2} \left(\frac{-1}{N^i} \middle| \frac{N^j}{N^2 h^{ij} - N^i N^j} \right),$$

and $\sqrt{-g} = N\sqrt{h}$.

- Intrinsic curvature: $R_{ij}^{(3)}$
- Extrinsic curvature:

$$K_{ij} \equiv \frac{1}{2N} \left(\dot{h}_{ij} - \nabla_i N_j - \nabla_j N_i \right) \equiv \frac{1}{N} E_{ij}.$$

- Four-dimensional Ricci scalar:

$$R = R^{(3)} + N^{-2} (E^{ij} E_{ij} - E^2) + \text{total derivative},$$

where $E \equiv h^{ij} E_{ij}$.

- Inflation action:

$$S = \frac{1}{2} \int d^4x \sqrt{h} N \left[R^{(3)} + N^{-2} (E^{ij} E_{ij} - E^2) + N^{-2} (\dot{\phi} - N^i \partial_i \phi)^2 - h^{ij} \partial_i \phi \partial_j \phi - 2V \right],$$

where $M_{\text{Pl}} \equiv 1$.

- Varying the action with respect to N and N^i , we find

$$\begin{aligned} R^{(3)} - 2V - h^{ij} \partial_i \phi \partial_j \phi - N^{-2} [E_{ij} E^{ij} - E^2 - (\dot{\phi} - N^i \partial_i \phi)^2] &= 0, \\ \nabla_i [N^{-1} (E_j^i - E \delta_j^i)] &= 0. \end{aligned}$$

These are constraint equations for N and N^i .

We will work in **comoving gauge**:

$$h_{ij} = a^2 e^{2\zeta(t, \mathbf{x})} \delta_{ij}, \quad N \equiv 1 + \alpha(t, \mathbf{x}), \quad N_i \equiv \partial_i \beta(t, \mathbf{x}).$$

Ex: Show that

$$R^{(3)} = -2a^{-2}e^{-2\zeta} [2\partial^2 \zeta + (\partial \zeta)^2],$$

$$E_{ij} = a^2 e^{2\zeta} (H + \dot{\zeta}) \delta_{ij} - \partial_{(i} N_{j)} + 2N_{(i} \partial_{j)} \zeta - N_k \partial_k \zeta \delta_{ij}.$$

Ex: Show that

$$\alpha = \frac{\dot{\zeta}}{H}, \quad \partial^2 \beta = -\frac{\partial^2 \zeta}{H} + a^2 \frac{\dot{\phi}^2}{2H^2} \dot{\zeta}.$$

Plugging these results into the action, expanding to second order, performing integrations by parts and using the background equations of motion, we get

$$S_2 = \int dt d^3x a^3 \varepsilon \left(\dot{\zeta}^2 - \frac{1}{a^2} (\partial_i \zeta)^2 \right).$$

- Note that $\mathcal{L}_2 \propto \varepsilon \implies \zeta$ is pure gauge in the dS limit ($\varepsilon \rightarrow 0$)
- Note that ζ has no mass term $\implies$ frozen superhorizon modes

In conformal time, we have

$$S_2 = \frac{1}{2} \int d\eta d^3x z^2 [(\zeta')^2 - (\partial_i \zeta)^2], \quad z \equiv a\sqrt{2\varepsilon}.$$

Defining $u \equiv z\zeta$, the equation of motion is

$$u'' - \nabla^2 u - \frac{z''}{z} u = 0 \quad \implies \quad u_k'' + \left(k^2 - \frac{z''}{z} \right) u_k = 0$$

Ex: Show that

$$\frac{z''}{z} = \frac{1}{\eta^2} \left[2 + 3 \left(\varepsilon + \frac{1}{2} \kappa \right) \right],$$

where $\kappa \equiv \varepsilon' / (\mathcal{H}\varepsilon)$.

The equation of motion becomes

$$u_k'' + \left(k^2 - \frac{\nu^2 - 1/4}{\eta^2} \right) u_k = 0, \quad \text{where} \quad \nu \equiv \frac{3}{2} + \varepsilon + \frac{1}{2}\kappa,$$

whose Bunch–Davies solution is

$$u_k(\eta) = \frac{\sqrt{\pi}}{2} \sqrt{-\eta} H_\nu^{(1)}(-k\eta) .$$

Ex: Show that

$$z(\eta) = z_*(\eta/\eta_*)^{\frac{1}{2}-\nu},$$

where $\eta_* = -k_*^{-1}$.

The dimensionless power spectrum of ζ is

$$\begin{aligned} \Delta_\zeta^2(k) &= \frac{k^3}{2\pi^2} \frac{|u_k(\eta)|^2}{z^2(\eta)} \\ &= \frac{k^3}{2\pi^2} \frac{1}{2\varepsilon_* a_*^2} (-k_*\eta)^{2\nu-1} \frac{\pi}{4} (-\eta) |H_\nu^{(1)}(-k\eta)|^2 . \end{aligned}$$

• In the late-time limit, we get

$$\lim_{k\eta \rightarrow 0} |H_\nu^{(1)}(-k\eta)|^2 \approx \frac{2}{\pi} (-k\eta)^{-2\nu} \quad \Longrightarrow \quad \Delta_\zeta^2(k) = \frac{1}{8\pi^2 \varepsilon_*} \frac{H_*^2}{M_{\text{Pl}}^2} (k/k_*)^{3-2\nu} .$$

• The scalar tilt is

$$n_s - 1 \equiv \frac{d \ln \Delta_\zeta^2}{d \ln k} = -2\varepsilon_* - \kappa_* ,$$

which was first measured by WMAP: $n_s = 0.965 \pm 0.004$.

3.4 Gravitational Waves

Consider tensor metric perturbations:

$$ds^2 = a^2(\eta) \left[-d\eta^2 + (\delta_{ij} + h_{ij})dx^i dx^j \right],$$

where $\partial^i h_{ij} = h^i_i = 0$.

- Expanding the Einstein–Hilbert action to second order gives

$$S = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g} R = \frac{M_{\text{Pl}}^2}{8} \int d\eta d^3x a^2 \left[(h'_{ij})^2 - (\nabla h_{ij})^2 \right] + \dots$$

- Use rotational symmetry to write $\mathbf{k} = (0, 0, k)$ and

$$\frac{M_{\text{Pl}}}{\sqrt{2}} a h_{ij} \equiv \begin{pmatrix} u_+ & u_\times & 0 \\ u_\times & -u_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

- The action becomes

$$S = \frac{1}{2} \sum_{\lambda=+,\times} \int d\eta d^3x \left[(u'_\lambda)^2 - (\nabla u_\lambda)^2 + \frac{a''}{a} u_\lambda^2 \right].$$

- The equation of motion for each polarization mode is

$$u''_k + \left(k^2 - \frac{a''}{a} \right) u_k = 0,$$

where the effective mass can be written as

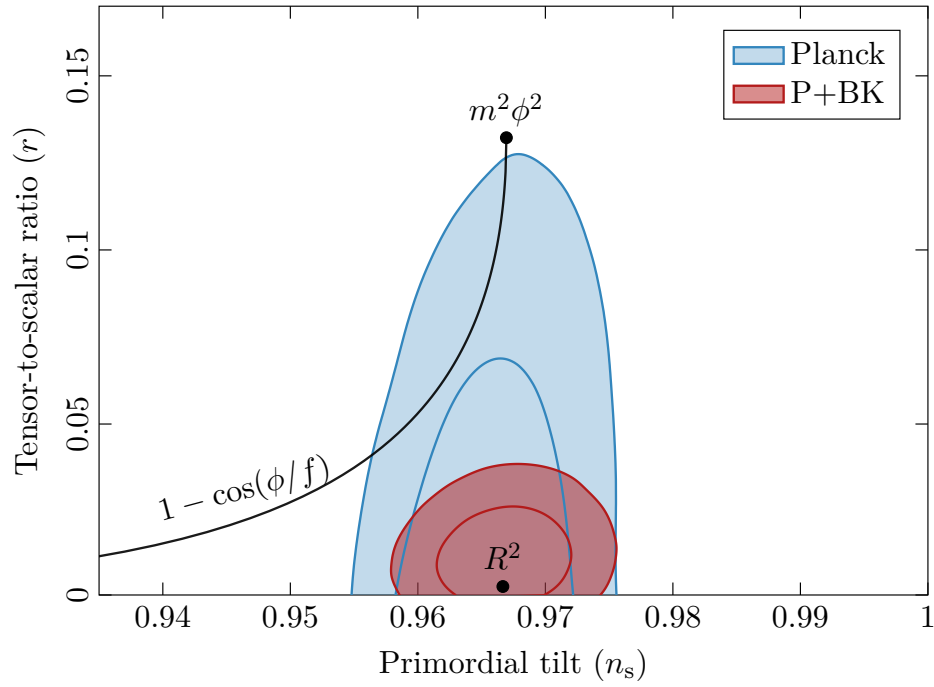
$$\frac{a''}{a} = \frac{\nu^2 - 1/4}{\eta^2}, \quad \text{with} \quad \nu \approx \frac{3}{2} + \varepsilon.$$

- Bunch–Davies mode function is the same as before.
- Superhorizon limit of the power spectrum:

$$\Delta_h^2(k) = 2 \times \left(\frac{2}{a M_{\text{Pl}}} \right)^2 \lim_{k\eta \rightarrow 0} \frac{k^3}{2\pi^2} |u_k(\eta)|^2 = \boxed{\frac{2}{\pi^2} \frac{H_*^2}{M_{\text{Pl}}^2} (k/k_*)^{3-2\nu}}.$$

- Observations are expressed in terms of the tensor-to-scalar ratio:

$$r \equiv \frac{\Delta_h^2(k_*)}{\Delta_\zeta^2(k_*)} = 16\varepsilon_*.$$



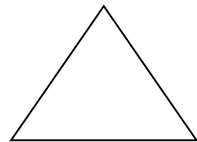
3.5 Primordial Non-Gaussianity

The main diagnostic of primordial non-Gaussianity is the **bispectrum**:

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle = B_\zeta(k_1, k_2, k_3) \times (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3).$$

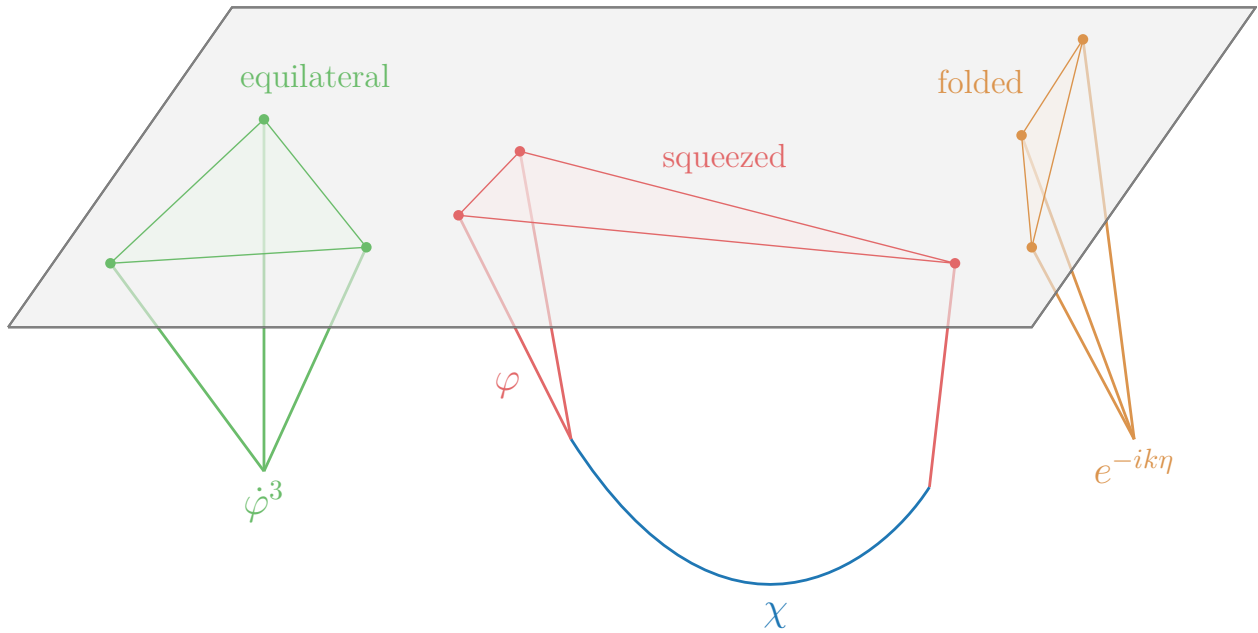
• amplitude: $f_{\text{NL}} \equiv \frac{5}{18} \frac{B_\zeta(k, k, k)}{P_\zeta^2(k)}$

• shape:


• effect: new particles new interactions excited states

• Planck constraints: $|f_{\text{NL}}^{\text{loc}}| < 5$ $|f_{\text{NL}}^{\text{equil}}| < 40$ $|f_{\text{NL}}^{\text{flat}}| < 20$



In the following, we describe three methods for computing these higher-point correlations:

- In-In Formalism
- Wavefunction Approach
- Cosmological Bootstrap

IV. IN-IN FORMALISM

In particle physics, the main observables are scattering amplitudes which predict the transition probabilities between some initial (“in”) and final (“out”) states (defined in terms free particles far in the past and future). In cosmology, however, there are no well-defined asymptotic regions where particles behave freely, which prevents a clean definition of scattering amplitudes. Instead, the primary observables are correlation functions, defined as expectation values at a fixed time in a quantum state evolved from an initial “in” state.

4.1 Master Formula

We are interested in computing

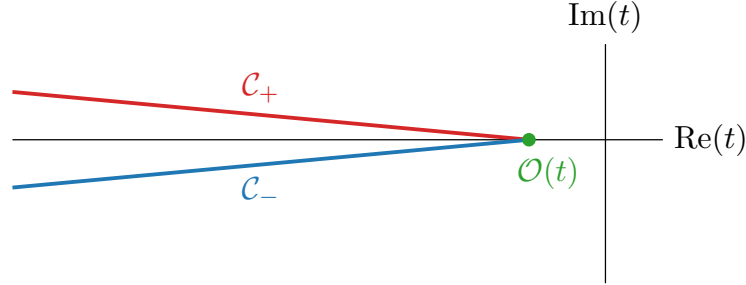
$$\langle \hat{\mathcal{O}}(t) \rangle \equiv \langle \Omega | \hat{\mathcal{O}}(t) | \Omega \rangle,$$

where $|\Omega\rangle$ the vacuum of the interacting theory.

In the interaction picture, we get

$$\boxed{\langle \hat{\mathcal{O}}(t) \rangle = \langle 0 | \bar{T} e^{i \int_{-\infty}^t dt' \hat{H}_{\text{int}}} \hat{\mathcal{O}}_I(t) T e^{-i \int_{-\infty}^t dt' \hat{H}_{\text{int}}} | 0 \rangle},$$

where H_{int} is the interaction Hamiltonian and the free vacuum $|0\rangle$ is selected by the “ $i\epsilon$ prescription”: $t \rightarrow t(1 - i\epsilon)$



In perturbation theory, we find

$$\langle \hat{\mathcal{O}}(t) \rangle = 2 \text{Im} \left(\int_{-\infty}^t dt' \langle 0 | \hat{\mathcal{O}}_I(t) \hat{H}_{\text{int}}(t') | 0 \rangle \right) + \dots$$

Let’s apply this to a few examples.

4.2 Some Examples

Consider the three-point function in ϕ^3 theory:

$$\begin{aligned} \langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \rangle' &\equiv \text{Diagram} \\ &= 2\lambda \text{Im} \left(f_{k_1}(t_*) f_{k_2}(t_*) f_{k_3}(t_*) \int_{-\infty}^{t_*} dt a^3(t) f_{k_1}^*(t) f_{k_2}^*(t) f_{k_3}^*(t) \right), \end{aligned}$$

where $f_k(t)$ is the mode function.

- In flat space, we have

$$f_k(t) = \frac{e^{-ikt}}{\sqrt{2k}}.$$

so that

$$\boxed{\langle \phi_1 \phi_2 \phi_3 \rangle' = -\frac{\lambda}{4k_1 k_2 k_3} \frac{1}{(k_1 + k_2 + k_3)}}.$$

- In de Sitter space, we have

$$\begin{aligned} f_k(\eta) &= H \sqrt{\frac{\pi}{4}} e^{i\frac{\pi}{4}(1+2\nu)} (-\eta)^{3/2} H_\nu^{(1)}(-k\eta), \quad \nu = \sqrt{\frac{9}{4} - \frac{m^2}{H^2}} \\ &= \begin{cases} (-H\eta) \frac{e^{-ik\eta}}{\sqrt{2k}} & m^2 = 2H^2 \\ \frac{H}{\sqrt{2k^3}} (1 + ik\eta) e^{-ik\eta} & m^2 = 0 \end{cases} \end{aligned}$$

so that

$$\langle \phi_{\mathbf{k}_1} \phi_{\mathbf{k}_2} \phi_{\mathbf{k}_3} \rangle' = \begin{cases} \frac{\pi}{8} \lambda \frac{H^2 \eta_*^3}{k_1 k_2 k_3}, \\ -\frac{\lambda H^2 (\gamma + \log(|K\eta_*|))}{12} \frac{\sum_i k_i^3 - 3K \sum_{i<j} k_i k_j + K^3}{(k_1 k_2 k_3)^3}, \end{cases}$$

where $K \equiv k_1 + k_2 + k_3$ and $\gamma = 2.7182 \dots$.

4.3 Slow-Roll Inflation

Consider

$$S = \int d^4x \sqrt{-g} \left(\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right).$$

Substituting

$$h_{ij} = a^2 e^{2\zeta(t, \mathbf{x})} \delta_{ij}, \quad N \equiv 1 + \frac{\dot{\zeta}}{H}, \quad \partial^i N_i \equiv -\frac{\partial^2 \zeta}{H} + a^2 \frac{\dot{\phi}^2}{2H^2} \dot{\zeta},$$

and expanding in powers of ζ , we find

$$S_2 = \int d\eta d^3x a^2 \varepsilon \left[(\zeta')^2 - (\partial_i \zeta)^2 \right],$$

$$S_3 = \int d\eta d^3x \left\{ a^2 \varepsilon^2 \left[\zeta (\zeta')^2 + \zeta (\partial_i \zeta)^2 - 2\zeta' \partial_i \zeta \partial_i^{-1} \zeta' \right] - \frac{1}{2} \partial_\eta (a^2 \varepsilon \kappa \zeta^2 \zeta') \right\}.$$

Exercise: Show that

$$\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle' = \frac{1}{32\varepsilon^2} \frac{H^4}{(k_1 k_2 k_3)^3} \left[\varepsilon \left(\sum_{i \neq j} k_i k_j^2 + \frac{8}{K} \sum_{i > j} k_i^2 k_j^2 \right) + (\kappa - \varepsilon) \sum_i k_i^3 \right].$$

Hint: The mode function is

$$f_k(\eta) \equiv \frac{H}{\sqrt{4\varepsilon k^3}} (1 + ik\eta) e^{-ik\eta}.$$

so that

$$\partial_\eta f^* \rightarrow \frac{H}{\sqrt{4\varepsilon k^3}} k^2 \eta e^{ik\eta},$$

$$(\partial_i f^*)^2 \rightarrow -(\mathbf{k}_1 \cdot \mathbf{k}_2) f_{k_1}^* f_{k_2}^* + \text{perms},$$

with $\mathbf{k}_1 \cdot \mathbf{k}_2 = \frac{1}{2}(k_3^2 - k_1^2 - k_2^2)$.

The amplitude of the bispectrum is

$$f_{\text{NL}} \equiv \frac{5}{18} \frac{B_\zeta(k, k, k)}{\Delta_\zeta^4(k)} = \boxed{\frac{5}{12} (9\varepsilon + \kappa)} \ll 1.$$

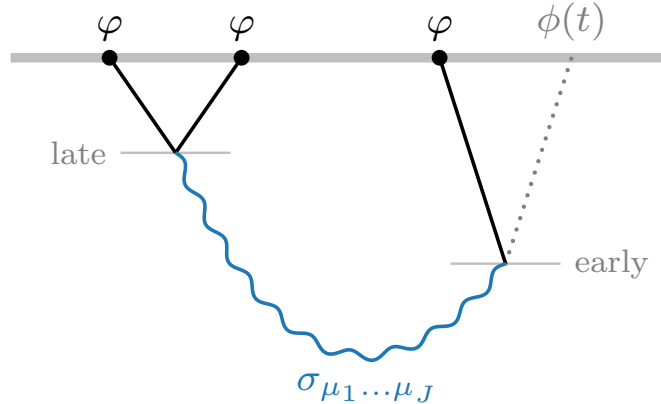
It is also interesting to take the *squeezed limit*, $k_3 \ll k_1 \approx k_2$, of the bispectrum:

$$\begin{aligned}
\lim_{k_3 \rightarrow 0} \langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle' &= \frac{1}{32\varepsilon^2} \frac{H^4}{k_1^6 k_3^3} \left[\varepsilon \left(2k_1^3 + \frac{8}{2k_1} k_1^4 \right) + (\kappa - \varepsilon) 2k_1^3 \right] \\
&= (2\varepsilon + \kappa) \left(\frac{H^2}{4\varepsilon} \frac{1}{k_1^3} \right) \left(\frac{H^2}{4\varepsilon} \frac{1}{k_3^3} \right) \\
&= (2\varepsilon + \kappa) P_\zeta(k_1) P_\zeta(k_3) \\
&= \boxed{(1 - n_s) P_\zeta(k_1) P_\zeta(k_3)} ,
\end{aligned}$$

where $P_\zeta(k)$ is the power spectrum. This result (called “single-field consistency relation”) applies to all models of single-field inflation, not just slow-roll models.

4.4 Cosmological Collider

New particles during inflation can lead to a violation of the single-field consistency relation:



Consider

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi)^2 - V(\phi) - \frac{1}{2}(\partial_\mu \sigma)^2 - \frac{1}{2}M^2 \sigma^2 + \frac{\sigma(\partial_\mu \phi)^2}{\Lambda} .$$

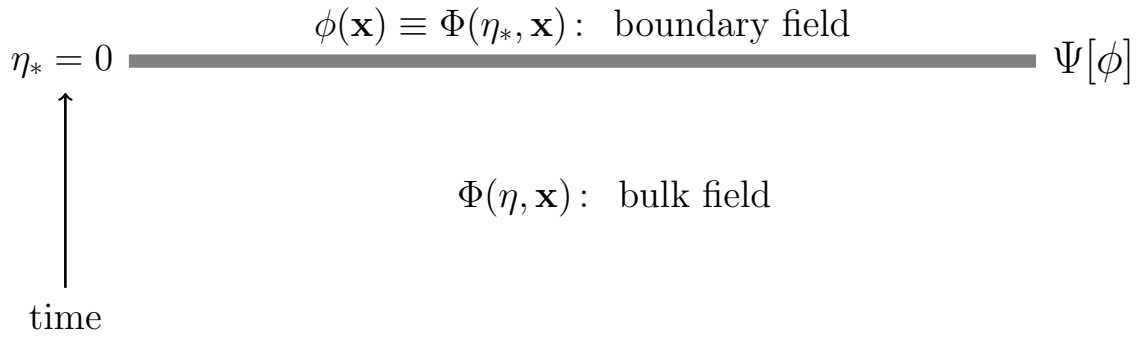
The associated bispectrum is hard to compute (see “cosmological bootstrap”). In the squeezed limit, we find

$$\boxed{\lim_{k_3 \rightarrow 0} \frac{\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle'}{P_\zeta(k_1) P_\zeta(k_3)} \sim \frac{\varepsilon}{8\pi} \frac{M_{\text{Pl}}^2}{\Lambda^2} e^{-\pi M/H} \left(\frac{k_3}{k_1} \right)^{3/2} \sin \left[\frac{M}{H} \log \left(\frac{k_3}{k_1} \right) \right]}$$

V. WAVEFUNCTION APPROACH

Cosmological correlations can also be computed by means of the so-called “wavefunction of the universe.” The wavefunction is a slightly more primitive object than correlators themselves—the relation between them is roughly the same as that between the S -matrix and scattering cross-sections. As a result, the wavefunction is somewhat simpler than correlators in certain ways that we will see.

5.1 Wavefunction of the Universe



The “[wavefunction of the universe](#)” is $\Psi[\phi] \equiv \langle \phi(\mathbf{x}) | \Omega \rangle$. It defines boundary correlators

$$\langle \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_N) \rangle = \int \mathcal{D}\phi \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_N) |\Psi[\phi]|^2.$$

The perturbative expansion of the wavefunction (in momentum space) is

$$\Psi[\phi] = \exp \left(- \sum_{N=2}^{\infty} \frac{1}{N!} \int d^3k_1 \cdots d^3k_N \Psi_N(\underline{\mathbf{k}}) \phi_{\mathbf{k}_1} \cdots \phi_{\mathbf{k}_N} \right),$$

where the “[wavefunction coefficients](#)” are

$$\begin{aligned}
 \Psi_N(\underline{\mathbf{k}}) &= (2\pi)^3 \delta(\mathbf{k}_1 + \cdots + \mathbf{k}_N) \psi_N(\underline{\mathbf{k}}) \\
 &= (2\pi)^3 \delta(\mathbf{k}_1 + \cdots + \mathbf{k}_N) \langle \underset{\uparrow}{O_{\mathbf{k}_1}} \cdots O_{\mathbf{k}_N} \rangle'.
 \end{aligned}$$

dual operators: $\phi \rightarrow O, \gamma_{ij} \rightarrow T_{ij}$

The relation between correlators and wavefunction coefficients is

$$\begin{aligned}\langle\phi\phi\rangle' &= \frac{1}{2\text{Re}\langle OO\rangle'} , \\ \langle\phi\phi\phi\rangle' &= \frac{2\text{Re}\langle OOO\rangle'}{\prod_{a=1}^3 2\text{Re}\langle O_a O_a\rangle'} , \\ \langle\phi\phi\phi\phi\rangle' &= \frac{\langle OOOO\rangle'}{(\langle OO\rangle')^4} + \frac{\langle OOX\rangle^3}{\langle XX\rangle\langle OO\rangle^4} .\end{aligned}$$

The wavefunction has the following path integral representation:

$$\Psi[\phi] = \int \mathcal{D}\Phi e^{iS[\Phi]} \approx e^{iS[\Phi_{\text{cl}}]} ,$$

$\begin{array}{ccc} \Phi(\eta_*)=\phi & & \uparrow \\ \Phi(-\infty^-)=0 & & \text{tree level} \end{array}$

where $-\infty^- \equiv -\infty(1 - i\epsilon)$. [Note: Opposite $i\epsilon$ to canonical quantization!]

To find the wavefunction, we therefore need to find the classical solution for the bulk field with the correct boundary conditions.

5.2 Warmup in Quantum Mechanics

Consider our old friend the **simple harmonic oscillator**:

$$S[\Phi] = \int dt \left(\frac{1}{2} \dot{\Phi}^2 - \frac{1}{2} \omega^2 \Phi^2 \right) .$$

- The classical solution (with the correct boundary conditions) is

$$\Phi_{\text{cl}}(t) = \phi e^{i\omega t} .$$

- The on-shell action becomes

$$\begin{aligned} S[\Phi_{\text{cl}}] &= \int_{t_i}^{t_*} dt \left[\frac{1}{2} \partial_t (\dot{\Phi}_{\text{cl}} \Phi_{\text{cl}}) - \frac{1}{2} \Phi_{\text{cl}} \underbrace{(\ddot{\Phi}_{\text{cl}} + \omega^2 \Phi_{\text{cl}})}_{=0} \right] \\ &= \frac{1}{2} \dot{\Phi}_{\text{cl}} \Phi_{\text{cl}} \Big|_{t=t_*} \\ &= \frac{i\omega}{2} \phi^2 . \end{aligned}$$

- The wavefunction then is

$$\Psi[\phi] \approx \exp(iS[\Phi_{\text{cl}}]) = \exp\left(-\frac{\omega}{2}\phi^2\right).$$

- The quantum variance of the oscillator therefore is

$$\boxed{\langle \phi^2 \rangle = \frac{1}{2\omega}}.$$

- In QFT, the same result applies for each Fourier mode:

$$\boxed{\langle \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \rangle' = \frac{1}{2\omega_k}},$$

where $\omega_k = \sqrt{k^2 + m^2}$.

To make this more interesting, consider a **time-dependent oscillator**:

$$S[\Phi] = \int dt \left(\frac{1}{2} \textcolor{red}{A}(t) \dot{\Phi}^2 - \frac{1}{2} \textcolor{blue}{B}(t) \Phi^2 \right).$$

- The classical solution is

$$\Phi_{\text{cl}} = \phi K(t), \quad \text{with} \quad \begin{aligned} K(0) &= 1 \\ K(-\infty) &\sim e^{i\omega t} \end{aligned}$$

- The on-shell action becomes

$$\begin{aligned} S[\Phi_{\text{cl}}] &= \int_{t_i}^{t_*} dt \left[\frac{1}{2} \partial_t (A \dot{\Phi}_{\text{cl}} \Phi_{\text{cl}}) - \frac{1}{2} \Phi_{\text{cl}} \underbrace{(\partial_t (A \dot{\Phi}_{\text{cl}}) + B \Phi_{\text{cl}})}_{=0} \right] \\ &= \frac{1}{2} A \dot{\Phi}_{\text{cl}} \Phi_{\text{cl}} \Big|_{t=t_*} \\ &= \frac{1}{2} A \phi^2 \partial_t \log K \Big|_{t=t_*}. \end{aligned}$$

- The wavefunction then is

$$\Psi[\phi] \approx \exp(iS[\Phi_{\text{cl}}]) = \exp\left(\frac{i}{2} (A \partial_t \log K) \Big|_* \phi^2\right),$$

which implies

$$|\Psi[\phi]|^2 = \exp\left(-\text{Im}(A \partial_t \log K) \Big|_* \phi^2\right) \implies \boxed{\langle \phi^2 \rangle = \frac{1}{2 \text{Im}(A \partial_t \log K) \Big|_*}}.$$

5.3 Free Fields in de Sitter

Consider a **massless field in de Sitter**:

$$S = \int d\eta d^3x a^2(\eta) [(\Phi')^2 - (\nabla\Phi)^2] \\ = \frac{1}{2} \int d\eta \frac{d^3k}{(2\pi)^3} \left[\frac{1}{(H\eta)^2} \Phi'_{\mathbf{k}} \Phi'_{-\mathbf{k}} - \frac{k^2}{(H\eta)^2} \Phi_{\mathbf{k}} \Phi_{-\mathbf{k}} \right],$$

which is the same as the time-dependent oscillator.

- The classical solution is

$$\Phi_{\text{cl}} = \phi K(\eta), \quad \text{with} \quad \begin{aligned} K(0) &= 1 \\ K(-\infty) &\sim e^{ik\eta} \end{aligned}$$

The function $K(\eta)$ is the *bulk-to-boundary propagator*.

- For a massless field, we have

$$K(\eta) = (1 - ik\eta)e^{ik\eta}, \\ \log K(\eta) = \log(1 - ik\eta) + ik\eta,$$

and hence

$$\begin{aligned} \text{Im}(A\partial_\eta \log K)|_{\eta=\eta_*} &= \frac{1}{(H\eta_*)^2} \text{Im} \left(\frac{-ik}{1 - ik\eta_*} + ik \right) \\ &= \frac{1}{(H\eta_*)^2} \text{Im} \left(\frac{k^2\eta_* + ik^3\eta_*^2}{1 + k^2\eta_*^2} \right) \xrightarrow{\eta_* \rightarrow 0} \boxed{\frac{k^3}{H^2}}, \end{aligned}$$

- The two-point function then is

$$\boxed{\langle \phi_{\mathbf{k}} \phi_{-\mathbf{k}} \rangle' = \frac{H^2}{2k^3}}.$$

The result for a massive field is derived in the lecture notes.

5.4 Anharmonic Oscillator

Consider the following **anharmonic oscillator**:

$$S[\Phi] = \int dt \left(\frac{1}{2} \dot{\Phi}^2 - \frac{1}{2} \omega^2 \Phi^2 - \frac{1}{3} \lambda \Phi^3 \right).$$

- The classical equation of motion is

$$\ddot{\Phi}_{\text{cl}} + \omega^2 \Phi_{\text{cl}} = -\lambda \Phi_{\text{cl}}^2.$$

- A formal solution is

$$\Phi_{\text{cl}}(t) = \phi K(t) + i \int dt' G(t, t') (-\lambda \Phi_{\text{cl}}^2(t')),$$

where

$$K(t) = e^{i\omega t},$$

$$G(t, t') = \frac{1}{2\omega} \left(e^{-i\omega(t-t')} \theta(t-t') + e^{i\omega(t-t')} \theta(t'-t) - e^{i\omega(t+t')} \right).$$

- Computing the on-shell action is now a bit more subtle.

As before, we first write

$$\begin{aligned} S[\Phi] &= \int_{t_i}^{t_*} dt \left[\frac{1}{2} \partial_t (\Phi \dot{\Phi}) - \frac{1}{2} \Phi (\ddot{\Phi} + \omega^2 \Phi) - \frac{\lambda}{3} \Phi^3 \right] \\ &= \frac{1}{2} \Phi \dot{\Phi} \Big|_{t=t_*} + \int dt \left[-\frac{1}{2} \Phi (\ddot{\Phi} + \omega^2 \Phi) - \frac{\lambda}{3} \Phi^3 \right]. \end{aligned}$$

Since

$$\begin{aligned} \lim_{t \rightarrow 0} G(t, t') &= 0, \\ \lim_{t \rightarrow 0} \partial_t G(t, t') &= -ie^{i\omega t'} \neq 0, \end{aligned}$$

the boundary term is

$$\begin{aligned} \frac{1}{2} \Phi_{\text{cl}} \dot{\Phi}_{\text{cl}} \Big|_{t=t_*} &= \frac{1}{2} \phi \left(i\omega \phi - i\lambda \int dt' (-ie^{i\omega t'}) \Phi_{\text{cl}}^2(t') \right) \\ &= \frac{i\omega}{2} \phi^2 - \frac{\lambda}{2} \phi \int dt' e^{i\omega t'} \Phi_{\text{cl}}^2(t'). \end{aligned}$$

The action then becomes

$$S[\Phi_{\text{cl}}] = \frac{i\omega}{2}\phi^2 - \frac{\lambda}{2}\phi \int dt e^{i\omega t} \Phi_{\text{cl}}^2 \\ + \int dt \left[-\frac{1}{2} \left(\phi e^{i\omega t} - i\lambda \int dt' G(t, t') \Phi_{\text{cl}}^2(t') \right) \left(-\lambda \Phi_{\text{cl}}^2(t) \right) - \frac{\lambda}{3} \Phi_{\text{cl}}^3 \right].$$

The terms linear in ϕ cancel.

- The final on-shell action is

$$S[\Phi_{\text{cl}}] = \frac{i\omega}{2}\phi^2 - \frac{\lambda}{3} \int dt \Phi_{\text{cl}}^3(t) - \frac{i\lambda^2}{2} \int dt dt' G(t, t') \Phi_{\text{cl}}^2(t') \Phi_{\text{cl}}^2(t) .$$

- We then write the classical solution as $\Phi_{\text{cl}}(t) = \sum \lambda^n \Phi^{(n)}(t)$, where

$$\Phi^{(0)}(t) = \phi e^{i\omega t}, \\ \Phi^{(1)}(t) = i \int dt' G(t, t') \left(-(\Phi^{(0)}(t'))^2 \right) \\ = i \int dt' G(t, t') \left(-\phi^2 e^{2i\omega t'} \right) = \frac{\phi^2}{3\omega^2} (e^{2i\omega t} - e^{i\omega t}) .$$

- With this, the wavefunction becomes

$$\Psi[\phi] \approx e^{iS[\Phi_{\text{cl}}]} = \exp \left(-\frac{\omega}{2}\phi^2 - \frac{\lambda}{9\omega}\phi^3 + \frac{\lambda^2}{72\omega^3}\phi^4 + \dots \right) .$$

From this, we can compute $\langle \phi^3 \rangle$, $\langle \phi^4 \rangle$, etc.

5.5 Wavefunction in Field Theory

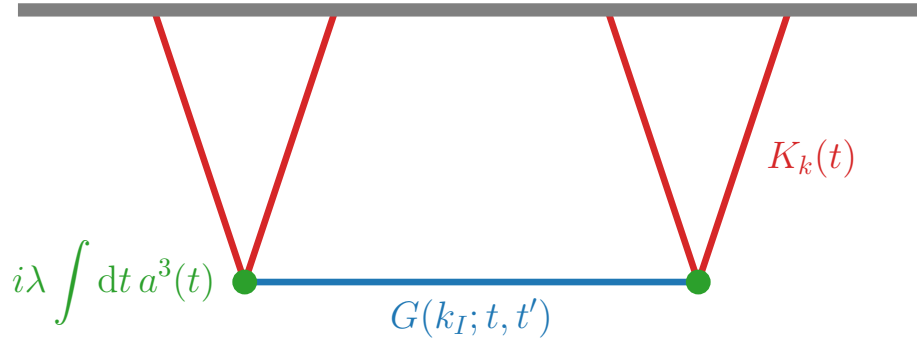
Back to field theory:

$$S[\Phi] = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi - \frac{1}{2} m^2 \Phi^2 - \frac{1}{3} \lambda \Phi^3 \right).$$

The analysis is similar to that of the anharmonic oscillator ($\Rightarrow$ lecture notes).

In the interest of time, we jump directly to **Feynman rules** for WF coefficients:

- bulk-to-boundary propagator K for every external line
- bulk-to-bulk propagator G for every internal line
- integrate each vertex over time.



Given a mode function $f_k(t)$, the bulk-to-boundary and bulk-to-bulk propagators are

$$K_k(t) = \frac{f_k(t)}{f_k(t_*)},$$

$$G(k; t, t') = \underbrace{f_k^*(t) f_k(t') \theta(t - t') + f_k^*(t') f_k(t) \theta(t' - t)}_{= G_F(k; t, t')} - \frac{f_k^*(t_*)}{f_k(t_*)} f_k(t) f_k(t').$$

5.6 Examples in Flat Space

The wavefunction is an interesting object even in flat space:

$$S = \int d^4x \left(-\frac{1}{2}(\partial\Phi)^2 - \frac{\lambda}{3!}\Phi^3 \right).$$

- Evaluate correlators at $t_* \equiv 0$.
- The flat-space mode function is

$$f_k(t) = \frac{1}{\sqrt{2k}} e^{ikt}.$$

- Using this, the relevant propagators are

$$K_k(t) = e^{ikt},$$

$$G(k; t, t') = \frac{1}{2k} \left(e^{-ik(t-t')} \theta(t-t') + e^{ik(t-t')} \theta(t'-t) - e^{ik(t+t')} \right).$$

We will compute the simplest tree-level correlators in this theory.

- The three-point wavefunction coefficient in Φ^3 theory is

$$\begin{aligned} \langle O_1 O_2 O_3 \rangle' &\equiv \text{Diagram: a triangle with a horizontal line at the top and a vertex at the bottom} \\ &= i\lambda \int_{-\infty}^0 dt e^{i(k_1+k_2+k_3)t} \\ &= \frac{\lambda}{k_1 + k_2 + k_3}. \end{aligned}$$

- This is easily generalized to N -point wavefunction coefficients:

$$\begin{aligned} \langle O_1 O_2 \dots Q_N \rangle' &= i\lambda \int_{-\infty}^0 dt e^{i(k_1+k_2+\dots+k_N)t} \\ &= \frac{\lambda}{k_1 + k_2 + \dots + k_N} \end{aligned}$$

- The four-point wavefunction coefficient in Φ^3 theory is

$$\begin{aligned}
\langle O_1 O_2 O_3 O_4 \rangle' &\equiv \text{Diagram: A trapezoid with a horizontal top line and a horizontal bottom line. Two diagonal lines connect the top corners to the bottom corners. The bottom corners are marked with black dots.} \\
&= -\lambda^2 \int_{-\infty}^0 dt dt' e^{ik_{12}t} G(k_I; t, t') e^{ik_{34}t'} \\
&= -\frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt dt' e^{ik_{12}t} \left(e^{-ik_I(t-t')} \theta(t-t') + e^{ik_I(t-t')} \theta(t'-t) - e^{ik_I(t+t')} \right) e^{ik_{34}t'} \\
&= -\frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt \int_{-\infty}^t dt' e^{i(k_{12}-k_I)t} e^{i(k_{34}+k_I)t'} \\
&\quad - \frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt' \int_{-\infty}^{t'} dt e^{i(k_{12}+k_I)t} e^{i(k_{34}-k_I)t'} \\
&\quad + \frac{\lambda^2}{2k_I} \int_{-\infty}^0 dt \int_{-\infty}^0 dt' e^{i(k_{12}+k_I)t} e^{i(k_{34}+k_I)t'} \\
&= \frac{\lambda^2}{2k_I} \left[\frac{1}{(k_{12}+k_{34})(k_{34}+k_I)} + \frac{1}{(k_{12}+k_{34})(k_{12}+k_I)} - \frac{1}{(k_{12}+k_I)(k_{34}+k_I)} \right] \\
&= \boxed{\frac{\lambda^2}{(k_{12}+k_{34})(k_{12}+k_I)(k_{34}+k_I)}}.
\end{aligned}$$

The answer has an interesting singularity structure. More about this later.

Exercise: Show that

$$\begin{aligned}
\langle O_1 O_2 O_3 O_4 O_5 \rangle' &\equiv \text{Diagram: A trapezoid with a horizontal top line and a horizontal bottom line. Two diagonal lines connect the top corners to the bottom corners. The bottom corners are marked with blue and red dots. A vertical line connects the top edge to the bottom edge, with a green dot on the bottom edge.} \\
&= \boxed{\frac{\lambda^3}{E_{\text{blue}} E_{\text{green}} E_{\text{red}}} \left[\frac{1}{k_{123} + k'_I} + \frac{1}{k_{345} + k_I} \right]}.
\end{aligned}$$

5.7 Examples in De Sitter

Consider a **massive scalar in de Sitter**.

- The equation of motion of $u = a(\eta)\Phi$ is

$$u'' + \left(k^2 + \frac{m^2/H^2 - 2}{\eta^2} \right) u = 0$$

- For generic m , the solution is

$$u_k(\eta) = \sqrt{\frac{\pi}{4}} e^{-\frac{i\pi}{4}(1+2\nu)} (-\eta)^{1/2} H_\nu^{(2)}(-k\eta).$$

- For $m^2 = 2H^2$, the field is conformally coupled and

$$u_k(\eta) = \frac{e^{ik\eta}}{\sqrt{2k}} \implies f_k(\eta) = \frac{u_k(\eta)}{a(\eta)} = H(-\eta) \frac{e^{ik\eta}}{\sqrt{2k}},$$

In the following, we consider this special case.

- The three-point wavefunction coefficient in Φ^3 theory is

$$\langle O_1 O_2 O_3 \rangle' = -\frac{i\lambda}{H^4(-\eta_*)^3} \int_{-\infty}^{\eta_*} \frac{d\eta}{\eta} e^{iK\eta} = \boxed{\frac{i\lambda}{H^4\eta_*^3} \log(-iK\eta_*)},$$

where $K = k_1 + k_2 + k_3$.

- The four-point wavefunction coefficient in Φ^3 theory is

$$\begin{aligned} \langle O_1 O_2 O_3 O_4 \rangle' &\equiv \text{Diagram: A horizontal line at the top with two vertical lines extending downwards from it, forming a trapezoid shape. The bottom two vertices are connected by a horizontal line, and the two slanted sides are also connected by a horizontal line, forming a rectangle. The bottom two vertices are marked with black dots.} \\ &= -\frac{\lambda^2}{H^6\eta_*^4} \int_{-\infty}^0 \frac{d\eta}{\eta} e^{ik_{12}\eta} \int_{-\infty}^0 \frac{d\eta'}{\eta'} e^{ik_{34}\eta'} \frac{1}{2k_I} \left[e^{-ik_I|\eta-\eta'|} - e^{ik_I(\eta+\eta')} \right]. \end{aligned}$$

Using

$$-i \int_a^\infty dx e^{ix\eta} = \frac{1}{\eta} e^{ia\eta},$$

this can be written as

$$\begin{aligned}
\langle O_1 O_2 O_3 O_4 \rangle' &= -\frac{\lambda^2}{H^6 \eta_*^4} \int_{k_{12}}^{\infty} dx \int_{k_{34}}^{\infty} dx' \int_{-\infty}^0 d\eta d\eta' e^{ix\eta} e^{ix'\eta'} G_{(\text{flat})}(k_I; \eta, \eta') \\
&= \frac{1}{H^6 \eta_*^4} \int_{k_{12}}^{\infty} dx \int_{k_{34}}^{\infty} dx' \langle O_1 O_2 O_3 O_4 \rangle'_{(\text{flat})}(x, x', k_I) \\
&= -\frac{\lambda^2}{H^6 \eta_*^4} \int_{k_{12}}^{\infty} dx \int_{k_{34}}^{\infty} dx' \frac{1}{(x+x')(x+k_I)(x'+k_I)} \\
&= \boxed{-\frac{\lambda^2}{H^6 \eta_*^4} \int_0^{\infty} dx \int_0^{\infty} dx' \frac{1}{(E+x+x')(E_L+x)(E_R+x')}}.
\end{aligned}$$

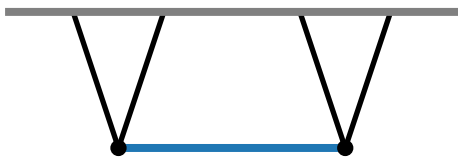
Evaluating the integrals gives

$$\begin{aligned}
\langle O_1 O_2 O_3 O_4 \rangle' &= \frac{\lambda^2}{2H^6 \eta_*^4 k_I} \left[\text{Li}_2 \left(\frac{k_{12} - k_I}{E} \right) + \text{Li}_2 \left(\frac{k_{34} - k_I}{E} \right) \right. \\
&\quad \left. + \log \left(\frac{E_L}{E} \right) \log \left(\frac{E_R}{E} \right) - \frac{\pi^2}{6} \right],
\end{aligned}$$

where Li_2 is the dilogarithm.

5.8 Cosmological Collider Physics

Consider the four-point function of a **conformally coupled scalar** (with $m^2 = 2H^2$) mediated by the exchange of a **massive scalar**:

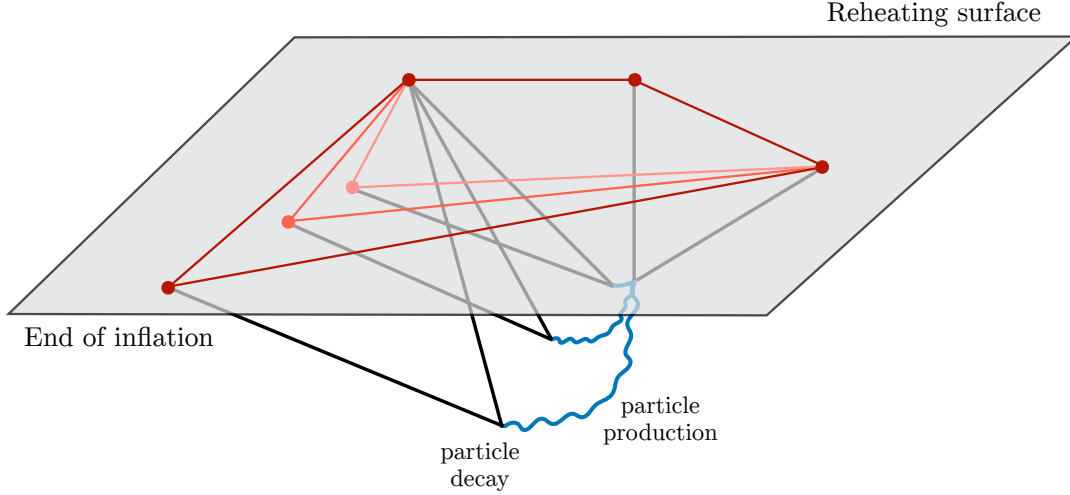
$$\begin{aligned}
F \equiv \langle O_1 O_2 O_3 O_4 \rangle &\equiv \text{Diagram} \\
&= -\lambda^2 \int \frac{d\eta}{\eta^2} \int \frac{d\eta'}{\eta'^2} e^{ik_{12}\eta} e^{ik_{34}\eta'} \underset{\substack{\uparrow \\ \text{products of Hankel functions}}}{G(k_I; \eta, \eta')}.
\end{aligned}$$


The time integrals cannot be performed analytically.

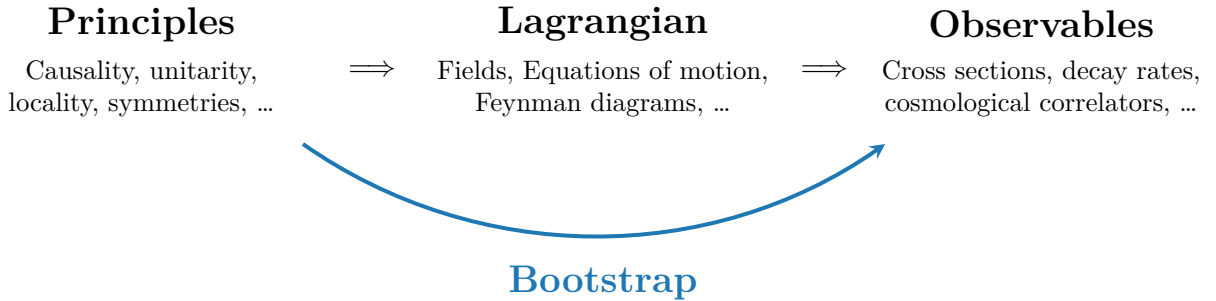
Is there a way to obtain an analytic understanding of this (and other) correlators in de Sitter space?

VI. COSMOLOGICAL BOOTSTRAP

In the previous two chapters, we presented two methods for computing inflationary correlations by evolving the fluctuations through the bulk spacetime and evaluating the resulting correlators on the future boundary. This involved complex time integrals (one for each vertex in a diagram) that in general cannot be computed analytically. Numerical integration is possible, but comes at the cost of masking the physical structure of the answers. In this chapter, we will describe a new approach to deriving the correlations directly on the boundary by imposing consistency with unitarity, locality and symmetry assumptions.



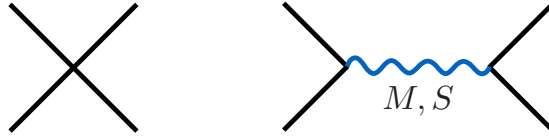
6.1 Bootstrap Philosophy



S-matrix Bootstrap

⇒ Scattering amplitudes are fixed by **Lorentz invariance**, **locality** and **unitarity**:

$$A(s, t) = \sum a_{nm} s^n t^m + \frac{\lambda^2}{s - M^2} P_s \left(1 + \frac{2t}{M^2} \right)$$



- No Lagrangian and Feynman diagrams are needed to derive this.
- Basic principles allow only a small menu of possibilities.

Cosmological Bootstrap

- What are the rules for consistent correlators?

Kinematics
Symmetries
Locality
Unitarity

- How are these rules implemented?

6.2 Symmetries

The metric of de Sitter space is

$$ds^2 = \frac{1}{H^2 \eta^2} (-d\eta^2 + d\mathbf{x}^2) .$$

- Besides translations and rotations, the metric is invariant under dilatations and special conformal transformations:

$$\begin{aligned} \text{D:} \quad & \eta \rightarrow \lambda \eta \\ & \mathbf{x} \rightarrow \lambda \mathbf{x} \\ \text{SCT:} \quad & \eta \rightarrow \frac{\eta}{1 + 2(\mathbf{b} \cdot \mathbf{x}) + b^2(x^2 - \eta^2)} \\ & \mathbf{x} \rightarrow \frac{\mathbf{x} + (x^2 - \eta^2)\mathbf{b}}{1 + 2(\mathbf{b} \cdot \mathbf{x}) + b^2(x^2 - \eta^2)} \end{aligned}$$

- The associated Killing vectors are

$$\begin{aligned}
P_i &= \partial_i , \\
J_{ij} &= x_i \partial_j - x_j \partial_i , \\
D &= -\eta \partial_\eta - x^i \partial_i , \\
K_i &= 2x_i \eta \partial_\eta + \left(2x^j x_i + (\eta^2 - x^2) \delta_i^j \right) \partial_j ,
\end{aligned}$$

which satisfy the de Sitter algebra $\text{so}(4, 1)$:

$$\begin{aligned}
[J_{ij}, J_{kl}] &= \delta_{jk} J_{il} - \delta_{ik} J_{jl} + \delta_{il} J_{jk} - \delta_{jl} J_{ik} , & [D, P_i] &= P_i , \\
[J_{ij}, K_l] &= \delta_{jl} K_i - \delta_{il} K_j , & [D, K_i] &= -K_i , \\
[J_{ij}, P_l] &= \delta_{jl} P_i - \delta_{il} P_j , & [K_i, P_j] &= 2\delta_{ij} D - 2J_{ij} .
\end{aligned}$$

- The infinitesimal transformations of a scalar field $\Phi(t, \mathbf{x})$ then are

$$\begin{aligned}
P_i \Phi &= \partial_i \Phi , \\
J_{ij} \Phi &= (x_i \partial_j - x_j \partial_i) \Phi , \\
D \Phi &= -(\eta \partial_\eta + x^i \partial_i) \Phi , \\
K_i \Phi &= 2x_i \eta \partial_\eta \Phi + \left(2x^j x_i + (\eta^2 - x^2) \delta_i^j \right) \partial_j \Phi .
\end{aligned}$$

How do the boundary fields $\phi(\mathbf{x})$ transform?

Consider a massive scalar field:

$$\Phi'' - \frac{2}{\eta} \Phi' - \nabla^2 \Phi + \frac{m^2}{H^2} \frac{1}{\eta^2} \Phi = 0 .$$

At late times, the solution is

$$\Phi(\mathbf{x}, \eta \rightarrow 0) = \phi(\mathbf{x}) \eta^\Delta + \bar{\phi}(\mathbf{x}) \eta^{3-\Delta} , \tag{*}$$

where

$$\Delta \equiv \frac{3}{2} - \sqrt{\frac{9}{4} - \frac{m^2}{H^2}} .$$

For “light fields”, with $m < \frac{3}{2}H$, the first term in (*) dominates at late times.

Using $\eta\partial_\eta \mapsto \Delta$, the boundary field profiles $\phi(\mathbf{x})$ transform as

$$\begin{aligned} P_i\phi &= \partial_i\phi, \\ J_{ij}\phi &= (x_i\partial_j - x_j\partial_i)\phi, \\ D\phi &= -(\Delta + x^i\partial_i)\phi, \\ K_i\phi &= [2x_i\Delta + (2x_ix^j\partial_j - x^2\partial_i)]\phi, \end{aligned}$$

like a primary operator of conformal weight Δ in a CFT.

The dual operator O transform in the same way, with $\Delta \mapsto 3 - \Delta \equiv \bar{\Delta}$.

How does this constrain the boundary correlators? $\implies$ **Ward identities:**

$$\begin{aligned} \text{D : } 0 &= \sum_{a=1}^N \left(\bar{\Delta}_a + x_a^j \frac{\partial}{\partial x_a^j} \right) \langle O_1 \cdots O_N \rangle, \\ \text{SCT : } 0 &= \sum_{a=1}^N \left(2\bar{\Delta}_a x_a^i + 2x_a^i x_a^j \frac{\partial}{\partial x_a^j} - x_a^2 \frac{\partial}{\partial x_{a,i}} \right) \langle O_1 \cdots O_N \rangle. \end{aligned}$$

In **Fourier space**, we have

$$\begin{aligned} DO &= [\bar{\Delta} + k^i \partial_{k^i}] O, \\ K_i O &= [2\bar{\Delta} \partial_{k^i} - k_i \partial_{k^j} \partial_{k^j} + 2k^j \partial_{k^j} \partial_{k^i}] O, \end{aligned}$$

and the Ward identity becomes

$$\begin{aligned} \left[-\textcolor{red}{3} + \sum_{a=1}^N D_a \right] \langle O_1 \cdots O_N \rangle'(\underline{\mathbf{k}}) &= 0, \\ \sum_{a=1}^N K_a^i \langle O_1 \cdots O_N \rangle'(\underline{\mathbf{k}}) &= 0. \end{aligned}$$

Dilatation symmetry fixes the overall momentum scaling:

$$\langle O_1 \cdots O_N \rangle' \propto k^{\bar{\Delta}_t - 3(N-1)}, \quad \text{where} \quad \bar{\Delta}_t \equiv \sum_{a=1}^N \bar{\Delta}_a.$$

All the juice is then in the SCT Ward identity.

Exercise: Show that the **two-point function** of two scalar operators is

$$\langle O_1 O_2 \rangle' \propto k_1^{2\bar{\Delta}_1 - 3} \delta_{\bar{\Delta}_1, \bar{\Delta}_2}.$$

Exercise: Consider the **three-point function** of two conformally coupled scalars Φ and a general scalar χ . Show that dilatation invariance implies

$$\langle O_1 O_2 X_3 \rangle' = k_3^{\bar{\Delta} - 2} \hat{G}(u),$$

where $u \equiv k_3/(k_1 + k_2)$, and that special conformal invariance requires

$$\left[u^2(1 - u^2) \partial_u^2 - 2u^3 \partial_u + \frac{m^2}{H^2} - 2 \right] \hat{G}(u) = 0.$$

This differential equation has a solution in terms of ${}_2F_1(u)$.

Four-point functions are less constrained by conformal symmetry alone. We need additional input to bootstrap them.

6.4 Singularities

Cosmological correlators have an interesting singularity structure.

We have seen this in our results for the **flat-space wavefunction**:

$$\begin{aligned}
 \langle O_1 O_2 O_3 \rangle &\equiv \text{Diagram: a horizontal line at the top with three lines descending from it to a single vertex at the bottom.} \\
 &= i\lambda \int_{-\infty}^0 dt e^{iKt} = \frac{\lambda}{\textcolor{blue}{K}}, \quad \text{where } K \equiv k_1 + k_2 + k_3. \\
 &\quad \uparrow \\
 &\quad \text{total energy singularity}
 \end{aligned}$$

$$\begin{aligned}
 \langle O_1 O_2 O_3 O_4 \rangle &\equiv \text{Diagram: a horizontal line at the top with two lines descending from it to two vertices, which are connected by a horizontal line.} \\
 &= \frac{\lambda^2}{\textcolor{blue}{E} \textcolor{red}{E}_L \textcolor{red}{E}_R}, \quad \text{where } \begin{aligned} E &\equiv k_{12} + k_{34}, \\ E_L &\equiv k_{12} + k_I, \\ E_R &\equiv k_{34} + k_I. \end{aligned} \\
 &\quad \uparrow \\
 &\quad \text{partial energy singularity}
 \end{aligned}$$

The residues of these singularities are **scattering amplitudes**:

$$\begin{aligned}
 \lim_{E \rightarrow 0} \frac{\lambda^2}{E E_L E_R} &= \frac{1}{E} \frac{\lambda^2}{(k_{12} + k_I)(-k_{12} + k_I)} = \frac{1}{E} \frac{\textcolor{red}{\lambda}^2}{\textcolor{red}{s}} = \frac{\textcolor{red}{A}_4}{E}, \\
 \lim_{E_L \rightarrow 0} \frac{\lambda^2}{E E_L E_R} &= \frac{\textcolor{red}{\lambda}}{E_L} \frac{\textcolor{blue}{\lambda}}{(\textcolor{blue}{k}_{34} + \textcolor{blue}{k}_I)(\textcolor{blue}{k}_{34} - \textcolor{blue}{k}_I)} = \frac{\textcolor{red}{A}_3 \times \tilde{\psi}_3}{E_L},
 \end{aligned}$$

where $\tilde{\psi}_3$ is the **on-shell wavefunction coefficient**:

$$\tilde{\psi}_3 \equiv \frac{1}{2k_I} \left[\psi_3(k_{34}, -k_I) + \psi_3(k_{34}, k_I) \right].$$

Generalization to de Sitter:

Wavefunctions factorize on partial energy singularities:

$$\boxed{\lim_{E_\Gamma \rightarrow 0} \psi_n = \frac{1}{E_\Gamma^p} \tilde{A}_\Gamma^{(m)} \times \tilde{\psi}_{\Gamma'}^{(n-m)}},$$

where

$$\begin{aligned} \tilde{A}_\Gamma &\equiv k_1^{\nu_1 - \frac{1}{2}} \cdots k_m^{\nu_m - \frac{1}{2}} A_\Gamma, \\ \tilde{\psi}_{\Gamma'} &\equiv P(k_I) \left(\psi_{\Gamma'}(k_a, -k_I) - \psi_{\Gamma'}(k_a, k_I) \right). \end{aligned}$$

Derived using: 1) Early-time limit of propagators, or 2) Unitarity cuts.

6.5 Cosmological Collider

We are now ready to return to the challenge of massive particle exchange:

The four-point function of conformally coupled scalars can be written as

$$\langle O_1 O_2 O_3 O_4 \rangle' = \begin{array}{c} \text{Diagram: A diamond shape with a vertical line from the top vertex to the bottom vertex. The left edge is blue and labeled k_2 at the top and k_1 at the bottom. The right edge is red and labeled k_3 at the top and k_4 at the bottom. The vertical line is labeled k_I in the middle.} \end{array} = \frac{1}{k_I} \hat{F}(\textcolor{blue}{u}, \textcolor{red}{v}),$$

where we have introduced

$$\textcolor{blue}{u} = \frac{k_I}{k_1 + k_2}, \quad \textcolor{red}{v} = \frac{k_I}{k_3 + k_4}.$$

This ansatz automatically solves the dilatation Ward identity.

After some work, the **conformal Ward identity** can be written as

$$\boxed{(\nabla_u - \nabla_v) \hat{F} = 0},$$

where $\Delta_u \equiv u^2(1 - u^2)\partial_u^2 - 2u^3\partial_u$ is our old friend.

We will classify solutions by their singularities.

Contact solutions

The simplest solutions correspond to contact interactions:

$$\hat{F}_c = \text{---}\bigvee = \sum_n \frac{c_n(u, v)}{\textcolor{red}{E}^{2n+1}},$$

$\uparrow$
 $\Phi^4, (\partial_\mu \Phi)^4, \dots$

which have poles at vanishing total energy

$$E \equiv \sum_n k_n = \frac{u+v}{uv} k_I.$$

Note that $F_c^{(n)} = \Delta_u^n F_c^{(0)}$, where $F_c^{(0)} \equiv uv/(u+v)$.

Exchange solutions

Next, we look for solutions corresponding to the tree-level exchange:

$$\hat{F}_e = \text{---}\bigwedge\text{---}$$

we try

$$\begin{aligned} (\Delta_u + M^2) \hat{F}_e &= \hat{F}_c, \\ (\Delta_v + M^2) \hat{F}_e &= \hat{F}_c, \end{aligned}$$

where $\hat{F}_c$ is a contact solution and $M^2 \equiv \mu^2 + \frac{1}{4}$.

Using the simplest contact interaction as a source, we have

$$\left[u^2(1-u^2)\partial_u^2 - 2u^3\partial_u + \left(\mu^2 + \frac{1}{4} \right) \right] \hat{F}_e = \lambda^2 \frac{uv}{u+v}. \quad (*)$$

Equation (*) has interesting **singularities**:

- Flat-space limit: $\lim_{u \rightarrow -v} \hat{F}_e = A_4 (u + v) \log(u + v)$
- Factorization limit: $\lim_{u, v \rightarrow -1} \hat{F}_e = A_3 \log(1 + u) \times A_3 \log(1 + v)$
- Folded limit: $\lim_{u \rightarrow +1} \hat{F}_e \propto \log(1 - u)$
- Collapsed limit: $\lim_{u \rightarrow 0} \hat{F}_e \propto u^{\frac{1}{2} + i\mu}$

Solution is fixed by: 1) No folded singularities and 2) Correct factorization.

Let's do it:

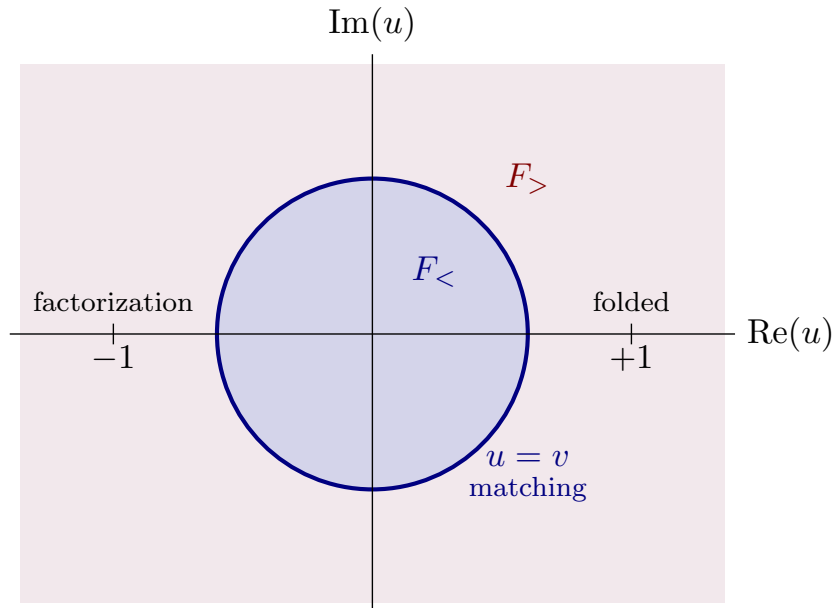
- The **homogeneous solution** of (*) is

$$\hat{F}_{\pm}(u) = \left(\frac{iu}{2\mu} \right)^{\frac{1}{2} \pm i\mu} {}_2F_1 \left[\begin{matrix} \frac{1}{4} \pm \frac{i\mu}{2}, \frac{3}{4} \pm \frac{i\mu}{2} \\ 1 \pm i\mu \end{matrix} \middle| u^2 \right],$$

and equivalent solutions $\hat{F}_{\pm}(v)$.

- To find the **particular solution**:

- 1) Find two series solutions around $u = 0$ and $u = \infty$.
- 2) Match these solutions at $u = v$:



3) Impose absence of folded singularities at $u = 1$ and $v = 1$.

$$\lim_{u \rightarrow 1} \hat{F}_{\pm}(u) = \alpha_{\pm}(\mu) \log(1 - u).$$

4) Demand correct factorization at $u = v = -1$:

$$\lim_{u, v \rightarrow -1} \hat{F} = A_3 \log(1 + u) \times A_3 \log(1 + v),$$

where $A_3 \equiv 1$.

The result is

$$\boxed{\hat{F}(u, v) = \sum_{m, n=0}^{\infty} c_{mn}(\mu) u^{2m+1} (u/v)^n + \frac{\pi}{2 \cosh \pi \mu} \hat{g}(u, v)} \quad u \leq v$$

where we have defined

$$c_{mn}(\mu) \equiv \frac{(-1)^n}{n} \prod_{i=0}^{2m} \frac{(n+i)}{(n + \frac{1}{2} + i)^2 + \mu^2}$$

and

$$\begin{aligned} \hat{g}(u, v) \equiv & (\beta - 1) \left[\hat{F}_+(u) \hat{F}_-(v) - \alpha \hat{F}_-(u) \hat{F}_-(v) \right] \\ & + (\beta + 1) \left[\hat{F}_-(u) \hat{F}_+(v) - \alpha^* \hat{F}_+(u) \hat{F}_+(v) \right], \end{aligned}$$

with

$$\begin{aligned} \beta &\equiv \frac{2}{i \sinh \pi \mu}, \\ \alpha &\equiv \frac{\alpha_+}{\alpha_-} = \left(\frac{i}{\mu} \right)^{2i\mu} \frac{\Gamma(\frac{1}{2} - i\mu) \Gamma(1 + i\mu)}{\Gamma(\frac{1}{2} + i\mu) \Gamma(1 - i\mu)}. \end{aligned}$$

In the soft limit, we have

$$\lim_{u, v \rightarrow 0} \hat{F}(u, v) = \left(\frac{uv}{4} \right)^{\frac{1}{2} + i\mu} (1 + i \sinh \pi \mu) \frac{\Gamma(\frac{1}{2} + i\mu)^2 \Gamma(-i\mu)^2}{2\pi} + \text{c.c.}$$

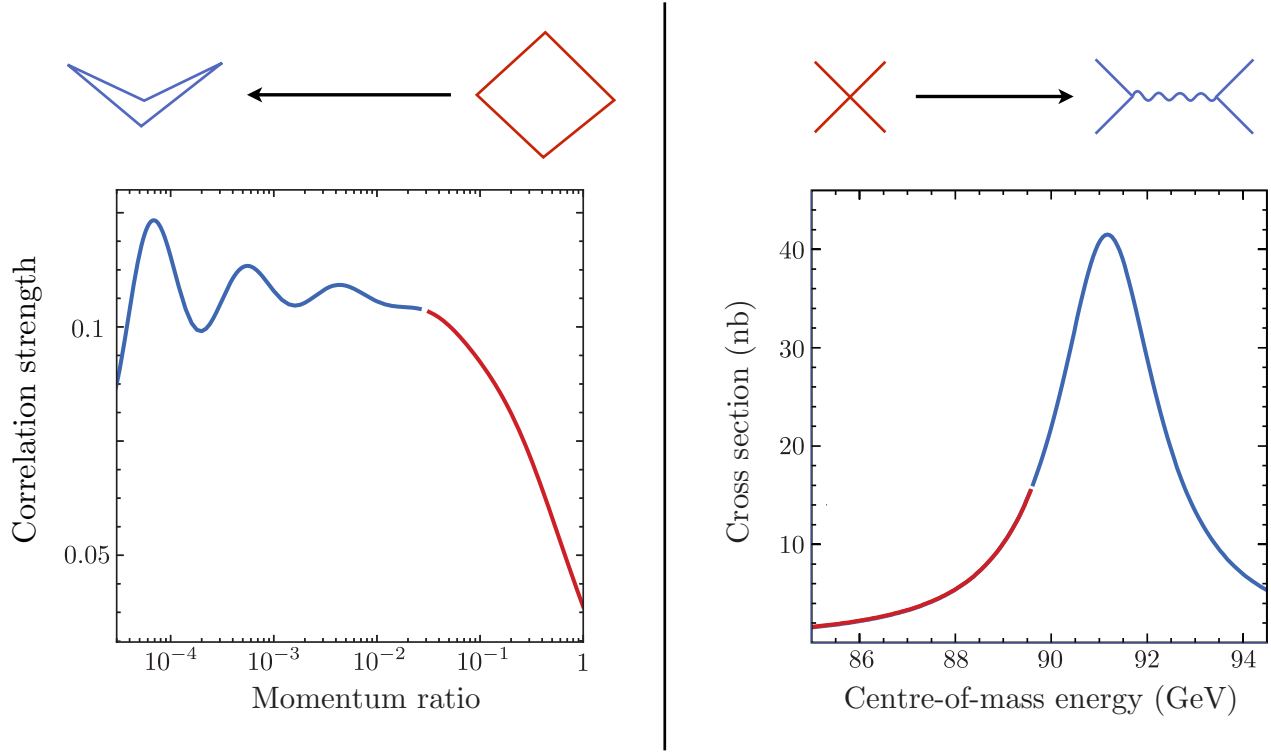
Summary:

The full solution has the form:

$$\hat{F}(u, v) = \hat{F}_{\text{EFT}}(u, v) + \hat{F}_{\text{pp}}(u, v) .$$

We are **forced** to have **particle production**.

This leads to **oscillations** in the collapsed limit:



These oscillations are the analog of a resonance in collider physics:

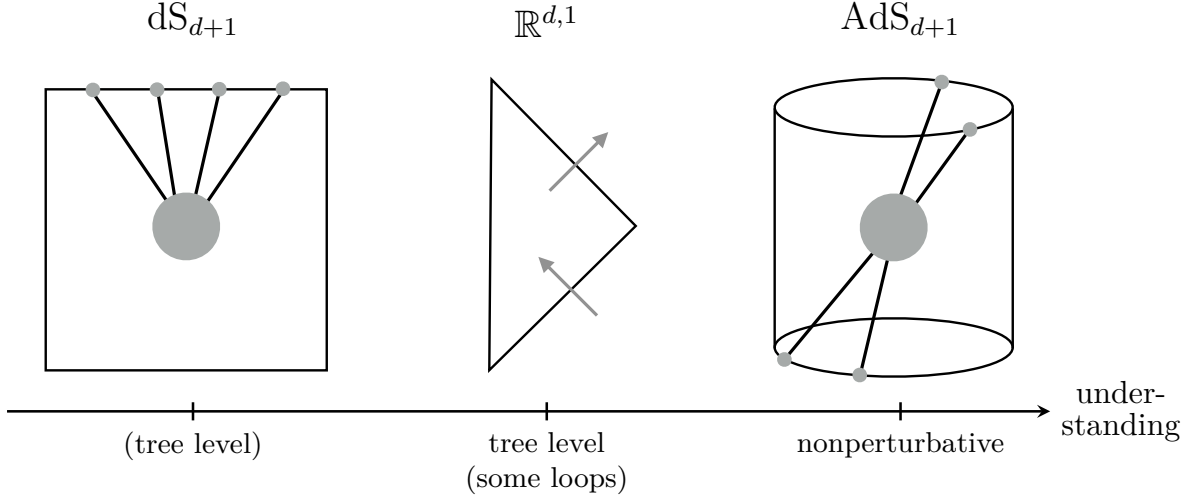
- The frequency of the oscillations depends on the **mass** of the particles.
- The angular dependence of the signal depends on the **spin** of the particles.

$$F_S = \text{[Diagram: A triangle with a blue line segment labeled } k_I \text{ and an angle labeled } \theta \text{]} \propto \sin[M \log k_I] P_S(\cos \theta)$$

VII. OUTLOOK

There are many interesting things I didn't have time to describe in the lectures. Some more can be found in the lecture notes.

The bootstrap approach is only the beginning of a systematic exploration of cosmological correlators:



Many open questions remain:

- **Beyond Feynman** So far, the cosmological bootstrap has only been applied to individual Feynman diagrams (which by themselves are unphysical). However, in the S-matrix bootstrap, the real magic is found for on-shell diagrams. What is the on-shell formulation of the cosmological bootstrap? What magic will it reveal?
- **Beyond Perturbation Theory** So far, the bootstrap constraints (singularities, unitarity cuts, etc.) are only formulated in perturbation theory and implemented only at tree level. In contrast, we understand the conformal bootstrap nonperturbatively. Recently, some preliminary progress was made to understand unitarity constraints in de Sitter nonperturbatively, but much work remains. A nonperturbative bootstrap is required if we want to understand the UV completion of cosmological correlators.
- **Beyond Spacetime** In quantum gravity, spacetime is an emergent concept. In cosmology, the notation of “time” breaks down at the Big Bang. What replaces it? The cosmological bootstrap provides a modest form of emergent time, but there should be a much more radical approach where space and time are outputs, not inputs. For scattering amplitudes, such a reformulation was achieved in the form of the *amplituhedron*. What is the analogous object in cosmology? How does the Big Bang arise from it?